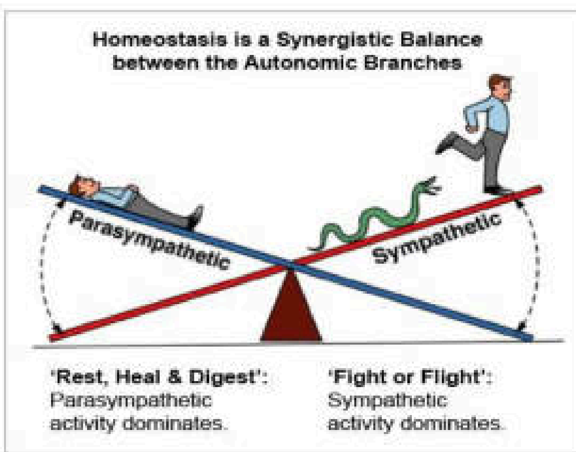
Sympathomimetic and Sympatholytic agents



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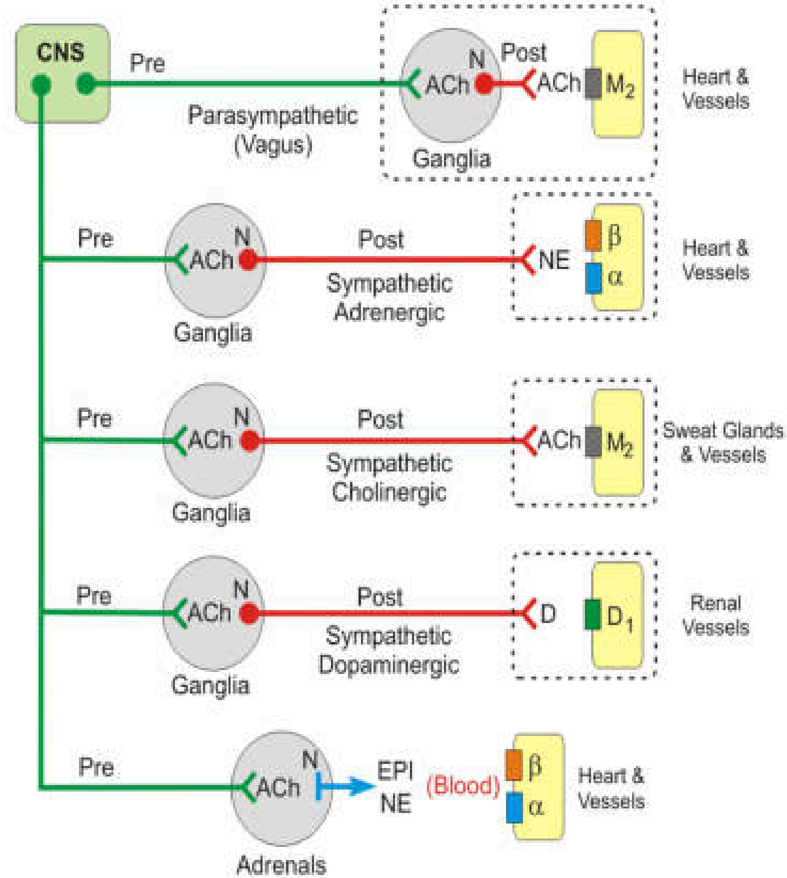
ภาควิชาเภสัชวิทยา

คณะเภสัชศาสตร์dมหาวิทยาลัยมหิดล

Pharmacology 1 (**Aug 27/28, 2568)**

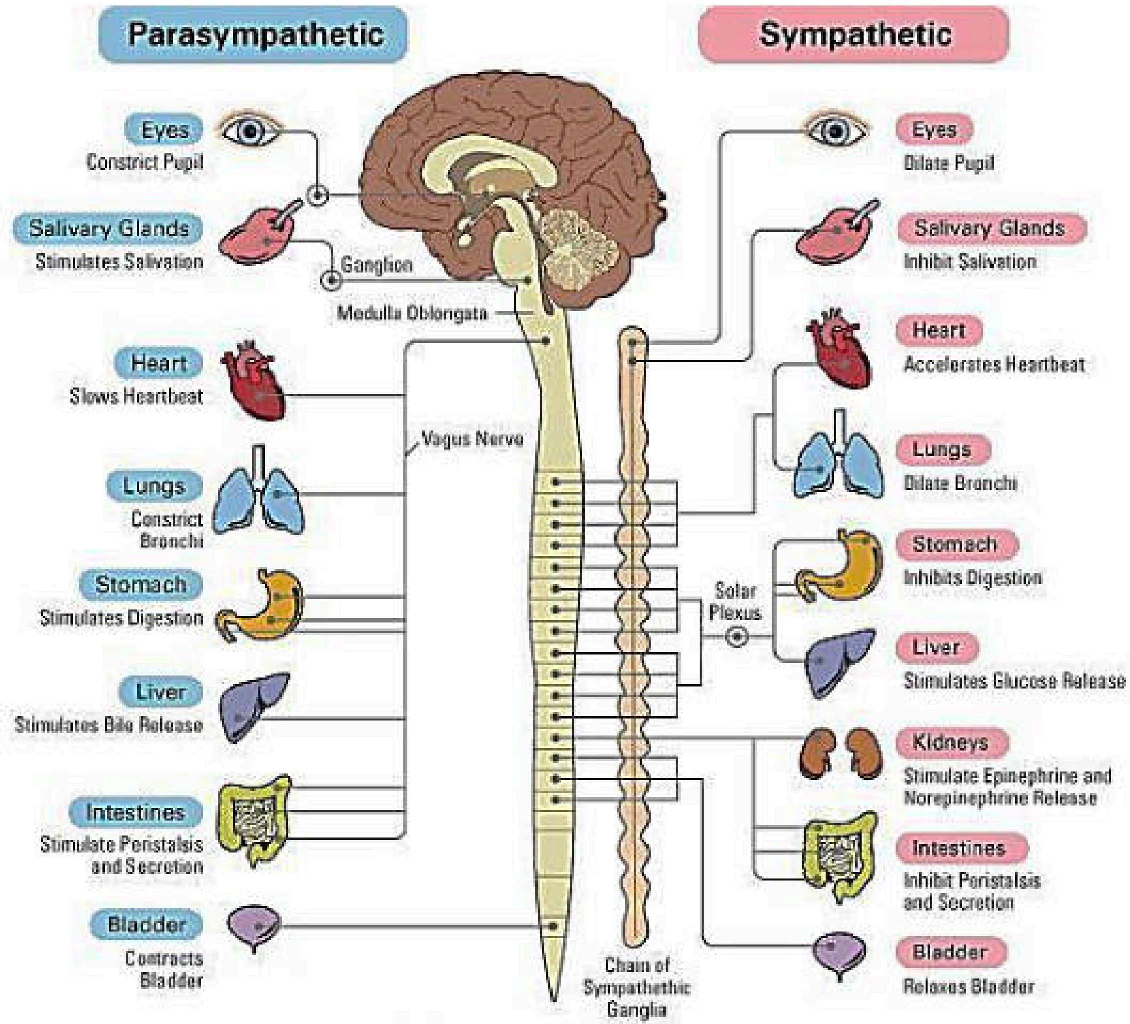
Sympathetic nervous system

- Control of **<u>CVS & Response to stress</u>**
- Neurotransmitters: **Norepinephrine (NE), Epinephrine (Epi)**
  - Epinephrine (adrenaline): synthesis predominates in adrenal medullary cells
  - NE: produce in the adrenergic neuron
- Act on Receptor called **"Adrenoceptor"**
- **"Sympathomimetic drugs (agents)":** Drugs causing sympathetic NE/Epi like action

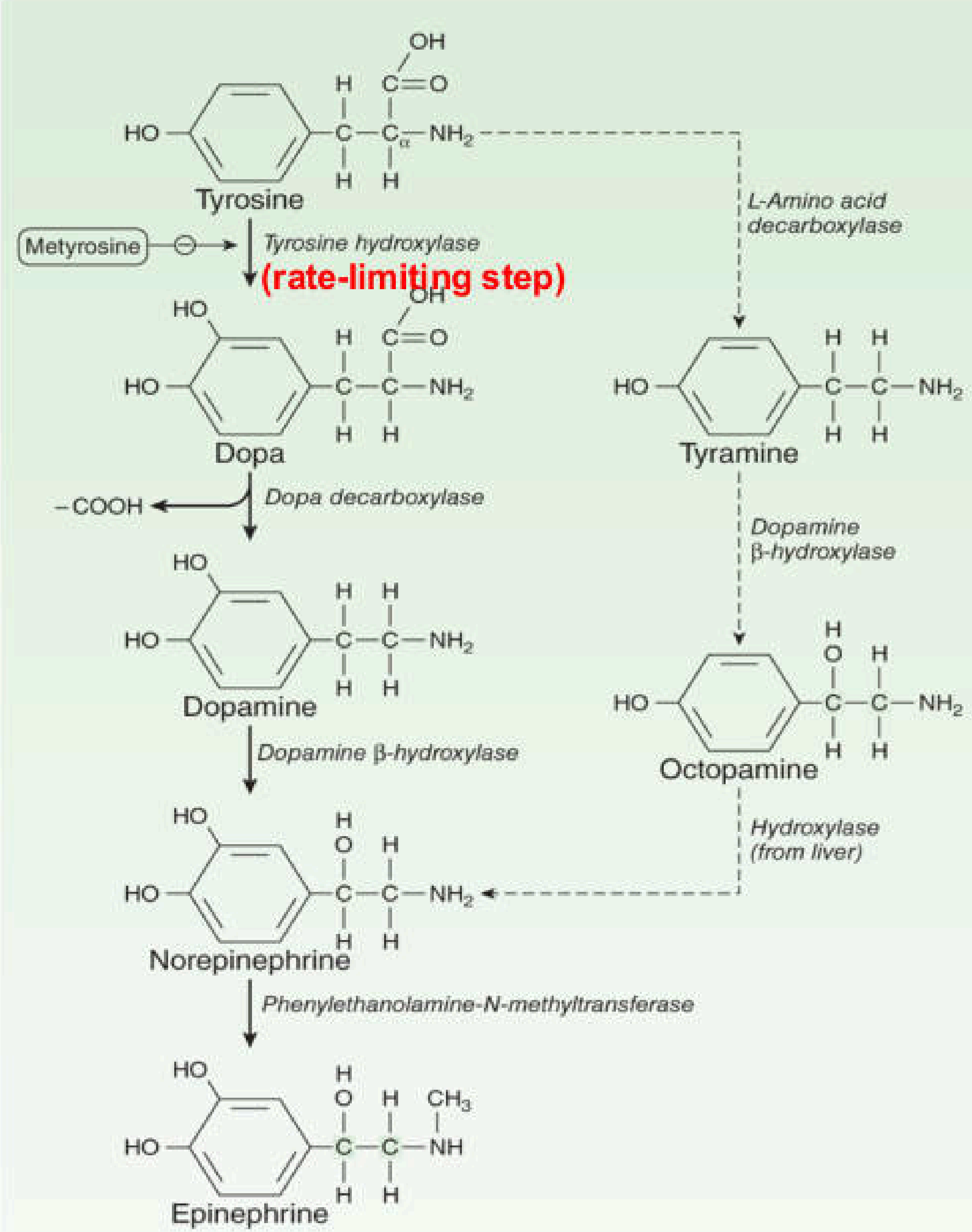


CNS = central nervous system; Pre = preganglionic; Post = postganglionic; ACh = acetylcholine; N = nicotinic receptor; NE = norepinephrine; EPI = epinephrine; D = dopamine; M_2 = muscarinic receptor; β = β-adrenoceptor; α = α-adrenoceptor; D_1 = dopaminergic receptor

Sympathetic nervous system functions



Schema Explaining How Parasympathetic and Sympathetic Nervous Systems Regulate Functioning Organs



Basic & Clinical Pharmacology, 14e; 2017

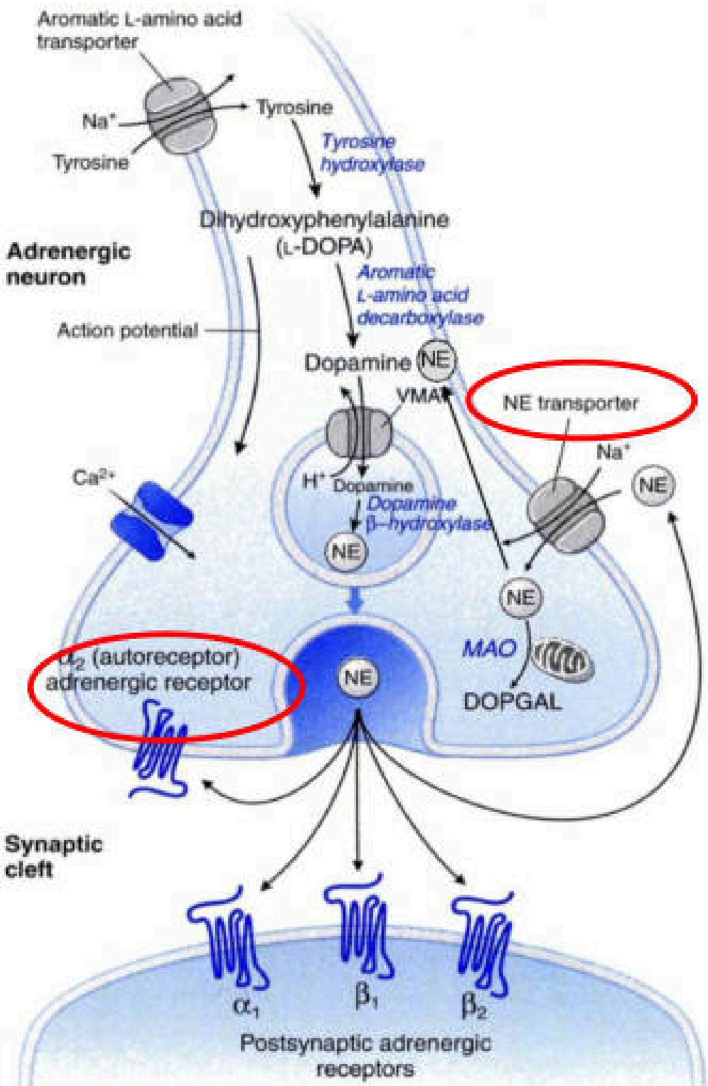
Biosynthesis of catecholamine

Catecholamines:

1. **Endogenous catecholamines**:
    - Epinephrine (Epi)
    - Norepinephrine (NE)
    - Dopamine (DA)

2. **Synthetic catecholamine**:
    - Isoproterenol (ISO)
    - Dobutamine

Steps of transmission



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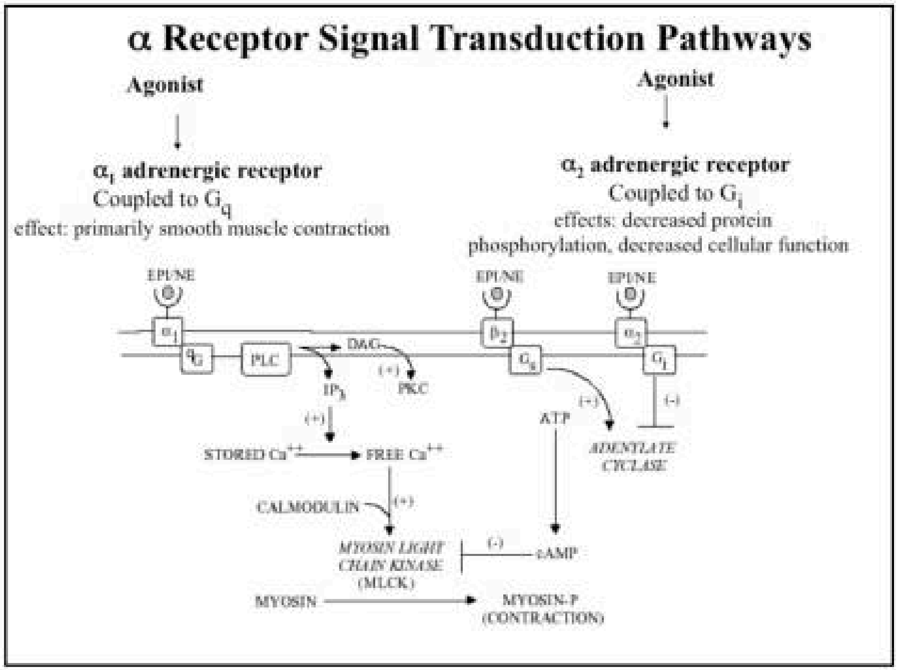
1. **Synthesis**: tyrosine
2. **Storage**
   -VMAT: NE contained in vesicle
3. **Release**
   -open voltage-gated Ca^{2+} channel
   - Ca^{2+} influx and exocytosis
4. NE **bind with receptors**
5. **Degradations**

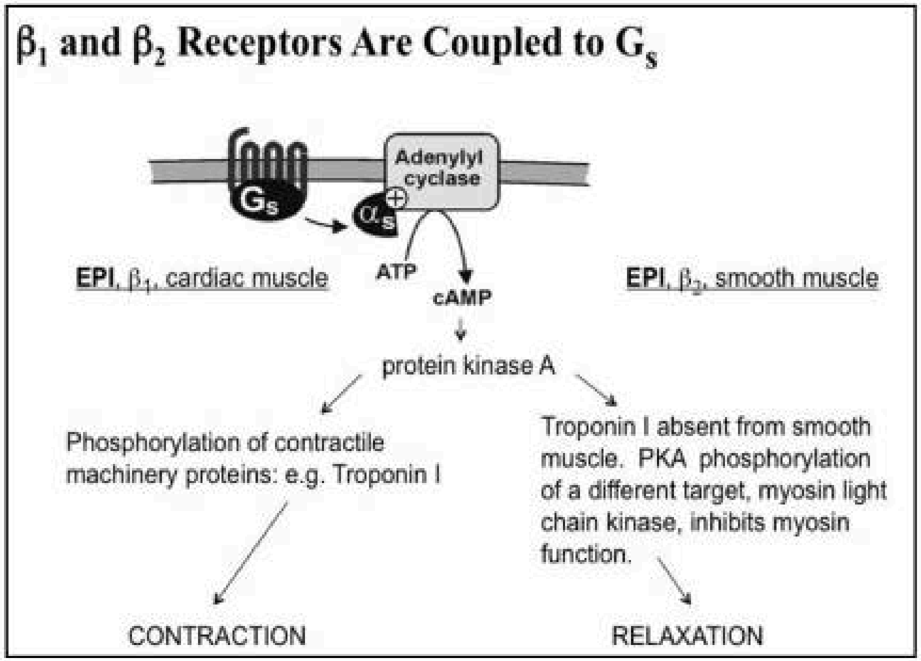
A. Uptake1: reuptake by recycled into the presynaptic nerve ending by *norepinephrine transporter* (NET)

B. Uptake 2: Bind to special inhibitory α2-adrenergic receptors on the presynaptic nerve ending → Autoreceptor → Decrease of NE release

C. Enzymes:
   A. Monoamine oxidase enzyme (MAO)
   B. Catecholamine –O- methyl transferase (COMT)

Adrenergic receptors





Dopamine receptors

D1-receptor

G_s protein

Activation of adenylyl cyclase

Locations: central nervous system, renal splanchnic, coronary vessel

Increase cAMP level

- Smooth muscle relaxation
- Vasodilation

- **D2-receptor**
- G_i protein
- Inhibition of adenylyl cyclase: Open K^+ channel, ↓ Ca^{2+} influx
- Locations: postsynaptic receptor in the striatum (plays an important role in dopaminergic neurotransmission and involved in motor control)

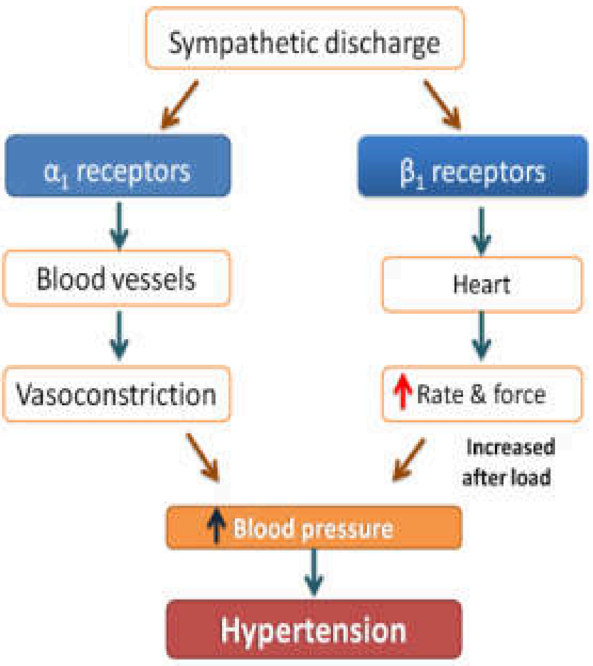
Receptor activities

Type	Tissue	Actions
α_1	Most vascular smooth muscle (innervated)	Contraction
	Pupillary dilator muscle	Contraction (dilates pupil)
	Pilomotor smooth muscle	Erects hair
	Prostate trigone, prostatic smooth muscle	Contraction
α_2	Adrenergic and cholinergic nerve terminals	Inhibits transmitter release
	Platelets	Stimulates aggregation
	Adrenergic and cholinergic nerve terminals	Inhibits transmitter release
	Some vascular smooth muscle	Contraction
	Adipocytes	Inhibits lipolysis
	Pancreatic β (B) cells	Inhibits insulin release

Type	Tissue	Actions
β_1	Juxtaglomerular cells	• increases renin release
	Heart	• Increases force and rate of heart contraction
β_2	Respiratory, uterine, and vascular smooth muscle	Smooth muscle relaxation
	Skeletal muscle	Promotes potassium uptake
	Human liver	Activates glycogenolysis
	Pancreatic β (B) cells	Stimulates insulin release
β_3	Bladder	Relaxes detrusor muscle
	Fat cells	Activates lipolysis
Dopamine1 (D1)	Smooth muscle	Dilates renal blood vessels
Dopamine2 (D2)	Nerve terminals	Inhibits adenylyl cyclase

Bertram G, Katzung, Basic & Clinical Pharmacology, 14e

Tests: pair the adrenergic receptor and its functional effects

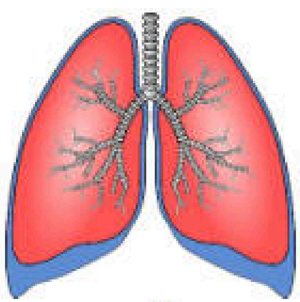


Sympathetic receptor	Location	Actions
1. α1 receptor	a. Vascular and bronchiolar smooth muscle	π Lipolysis, uterine relaxation
2. α2 receptor	b. Renal splanchnic, cerebral	♣ Bronchodilation
3. β1 receptor	c. Fat cells and detrusor muscle of the bladder	♥ Decrease sympathetic outflow
4. β2 receptor	d. CNS (presynaptic neuron)	♠ Relax renal vascular smooth muscle
5. β3 receptor	e. Cardiac cells	* Vascular smooth muscle contraction
6. D1 receptor	f. Vascular smooth muscle	♦ Increase heart rate and contractility, ↑Renin

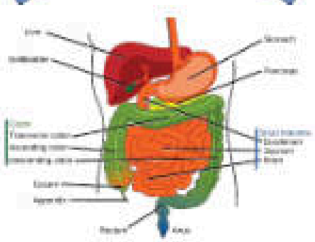
Most actions of sympathomimetic agents

Amphetamine, cocaine (enter the CNS): Stimulant CNS effects, reduction of fatigue, anorexia, euphoria, insomnia, seizures (high dose)

Phenylephrine: Mydriasis
Tetrahydrozoline: reduces the conjunctival itching and congestion caused by irritation or allergy
Alpha2-selective agonists (brimonidine, apraclonidine): reducing synthesis of aqueous humor for treatment of glaucoma



β2 agonists (albuterol, metaproterenol, terbutaline): relaxes the smooth muscle of the bronchi



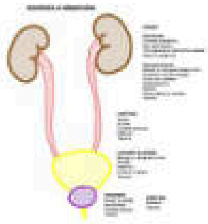
Gastrointestinal Tract

Activation of either α or β receptors leads to relaxation of the smooth muscle

Katzung & Trevor's Pharmacology: Examination & Board Review, 13ed

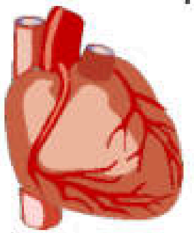
Most actions of sympathomimetic agents (cont.)

Genitourinary Tract



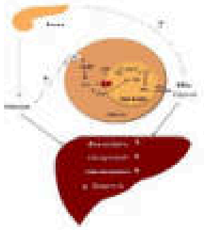
α1 receptors: prostatic smooth muscle contraction
Beta2 agonists (ritodrine, terbutaline): uterine relaxation

Heart



β1 receptor: increased rate of cardiac pacemakers, atrioventricular node conduction velocity, cardiac force
Epinephrine: acute cardiac stimulation
Norepinephrine: shock
Dobutamine: acute heart failure to increase cardiac output

Metabolic and Hormonal Effects



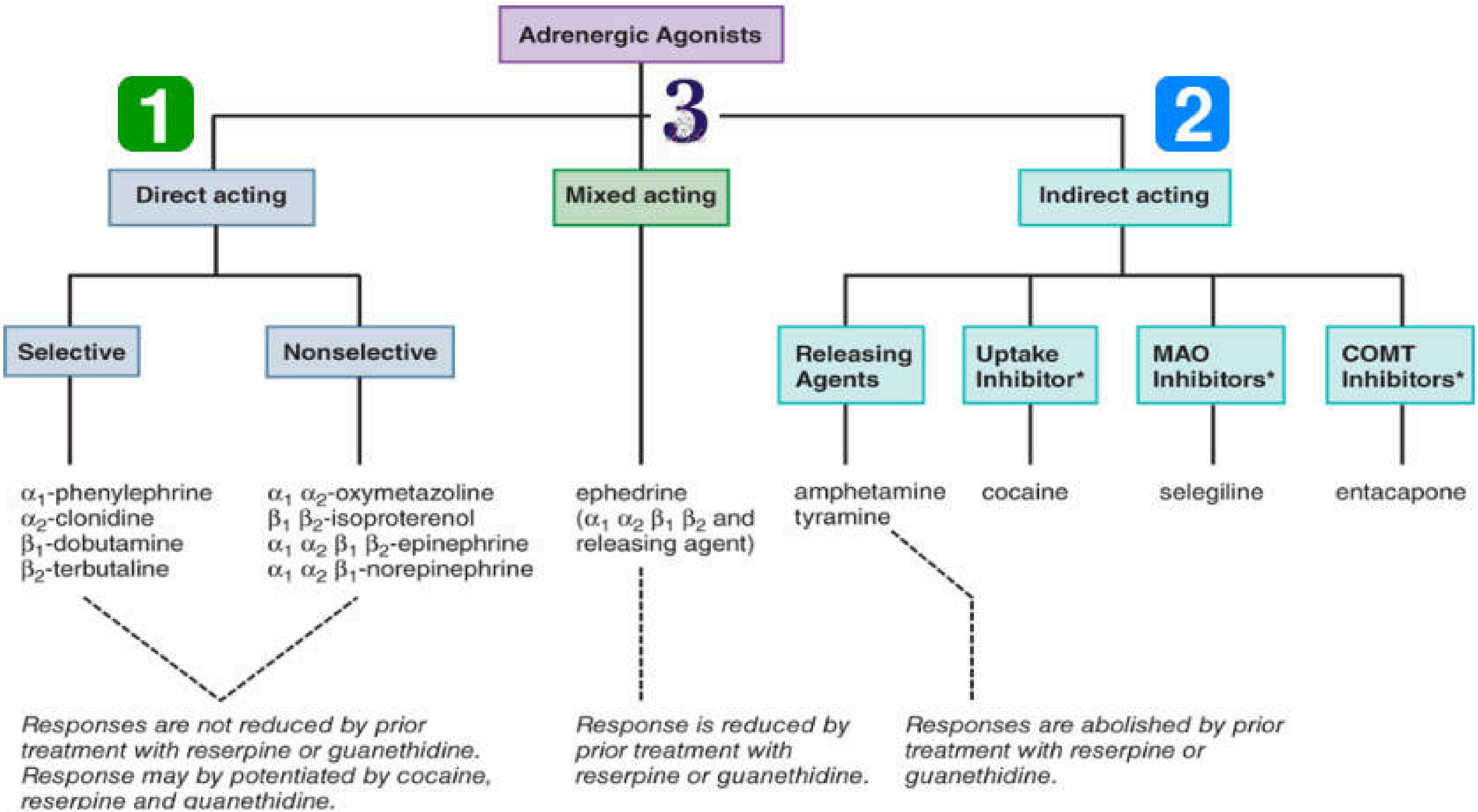
Beta1 agonists: increase renin secretion
Beta2 agonists: increase insulin secretion

Others:

Epinephrine: local hemostatic
Phenylephrine: nasal decongestant

Katzung & Trevor's Pharmacology: Examination & Board Review, 13e

Classification of sympathomimetic drugs



. Basic & Clinical Pharmacology, 14e; 2017

Non-selective Direct acting sympathomimetic drugs

1. **Epinephrine (Adrenaline)**
   - Activation of **$\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$ –receptors**
   - **Low dose epinephrine (0.01 µg/kg/min)**: predominately β receptor $\rightarrow$ + $\beta 2$ receptor $\rightarrow$ vasodilation $\rightarrow$ decreases BP
   - **High dose epinephrine (>0.2 µg/kg/min):** predominately both α and β receptors $\rightarrow$ increase BP
     - **Potent vasoconstrictor** ($\alpha 1$-receptor)
     - **Cardiac stimulant** ($\beta 1$-receptor): Positive inotropic, Positive chronotropic
     - Dose of continuous intravenous infusion 0.05-1 µg/kg/min for treatment of hypotension

- Quick degradation by MAO & COMT: short $T_{1/2}$:~2 min
- Cannot be orally absorbed, use only IV, SC
- **Adverse effects**: cerebral hemorrhage, cardiac arrhythmias, restlessness, throbbing headache, tremor, and palpitations

Indications of **epinephrine**

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1. **<u>Hypersensitivity reactions</u>**: drug of choice for <u>anaphylactic shock</u> (1-4 µg/min)
   - Epi (1:1000) 1 ml dilute with 5% D/W 250 ml and infuse with 15-60 ml/min (adjust according to patient BP)
2. **Restoring cardiac rhythm in patients with cardiac arrest**
3. **Prolong the action of local anesthetics**: Vasoconstriction
4. **Topical hemostatic agent on bleeding surfaces**
5. **Glaucoma (open-angle glaucoma):** Dipivefrin (New drug)

2.7.3. Drugs used in cardiopulmonary resuscitation				
1	Epinephrine	Adrenaline	sterile sol	ก

	ชื่อยา Generic name	ชื่อพ้องของยา Synonym	รูปแบบ Form, Strength	บัญชี
2	Lidocaine hydrochloride		gel, oint, spray, sterile sol, sterile sol (dental cartridge), viscous sol	ก
3	Lidocaine + Prilocaine		cream	ก
4	Lidocaine hydrochloride + Epinephrine		sterile sol, sterile sol (dental cartridge)	ก
5	Mepivacaine hydrochloride		sterile sol (dental cartridge)	ก
6	Mepivacaine hydrochloride + Epinephrine		sterile sol (dental cartridge)	ก
7	Bupivacaine hydrochloride		sterile sol	ง
8	Bupivacaine hydrochloride with/without glucose		sterile sol	ค
9	Lidocaine hydrochloride + Epinephrine		sterile sol	ค

12.3.5. Other dental preparations				
1	Epinephrine	Adrenaline	sterile sol	ก
2	Artificial saliva	Saliva substitutes	sol (hosp)	ก
3	Hydrogen peroxide		mouthwash sol (1.5% w/v)	ก
4	Sodium chloride		sterile sol	ก

2. Norepinephrine (NE or noradrenaline)

- Activation α (α1, α2) and β1-receptor (less effect to β2-receptor)
- Increase peripheral resistance, systolic BP & diastolic BP
  - Activate vagal reflex >>> negative chronotropic, decrease positive chronotropic effect
- Ineffective when given orally, rapidly inactivated in the body by COMT and MAO ($T_{1/2}$: ~ 3 min)
- **Therapeutic dose:** 0.02-2 μg/kg/min or 1-50 μg/min (intravenous infusion) for treatment of cardiogenic shock
- **Untoward effects**: similarto epinephrine
- **Therapeutic Uses:** Vasoconstrictor to raise or support blood pressure

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2.7.2. Vasoconstrictor sympathomimetics				
1	Norepinephrine	Noradrenaline	sterile sol (as bitartrate or hydrochloride)	ก
2	Ephedrine hydrochloride วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2		sterile sol	ค
3	Midodrine hydrochloride ยากำพร้า		tab	ค
4	Phenylephrine hydrochloride		sterile sol (50 mcg/ml)	ค

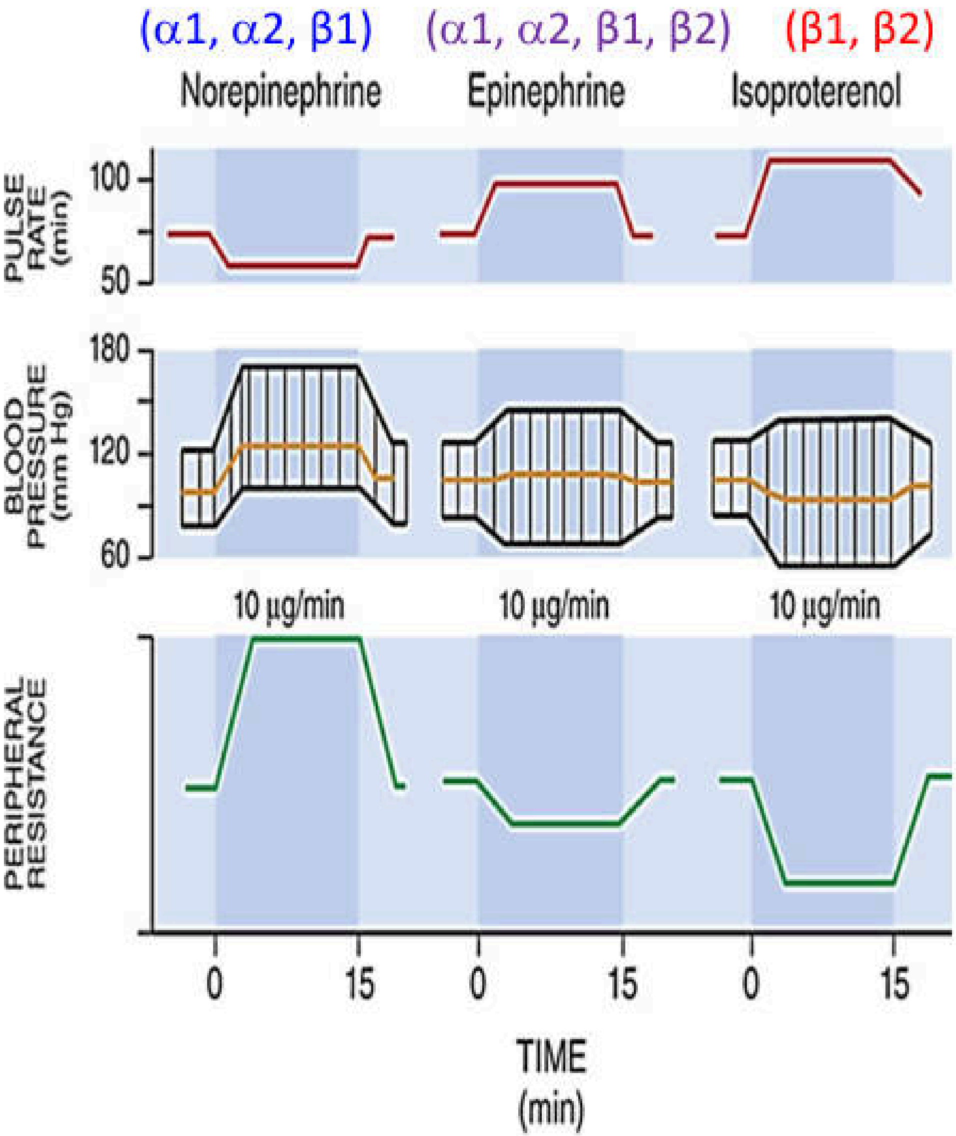
3. Isoproterenol (Isoprenaline)

Non-selective β-agonists

- β1 and β2 -agonists
- Effects on heart: Positive chronotropic, Positive inotropic, Increased cardiac output

Vasodilator and relax smooth muscle effects more than NE, Epinephrine (β2 receptor activation)

Use for **Inotropic sympathomimetic agents**



4. Non-selective alpha-receptor agonists:

Drugs: Oxymetazoline, xylometazoline, tetrahydrozoline

Drugs	Dosage forms	Indications
Antazoline hydrochloride + **Tetrahydrozoline** hydrochloride	eye drop	Treatment of eye allergic
Dexamethasone sodium phosphate + Chloramphenicol + **Tetrahydrozoline** hydrochloride	eye drop	Treatment of eye infection
Oxymetazoline hydrochloride	nasal drop, nasal spray	Topical nasal decongestants

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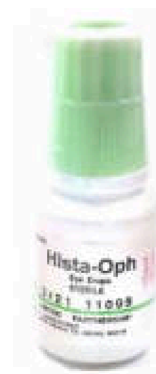
2.	Dexamethasone sodium phosphate + Chloramphenicol + Tetrahydrozoline hydrochloride	eye drop	ค

12.2.2 Topical nasal decongestants

1.	Ephedrine hydrochloride *วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2*	nasal drop (hosp) (เฉพาะ 0.5-3%)	ก
2.	Sodium chloride	sterile sol (for irrigation) (เฉพาะ 0.9%)	ก
3.	Oxymetazoline hydrochloride	nasal drop, nasal spray	ข



5. Dopamine

precursor of NE and Epi → activate both dopaminergic & adrenergic receptor

Low dose (1-3 µg/kg/min): stimulate **dopamine 1 receptor** >>> Vasodilation increase renal blood flow

Moderate dose (3-10 µg/kg/min): activate: beta-1 receptor: inotropic drugs

High dose (>10 µg/kg/min): α1+ β1-receptor activation: vasoconstriction, + inotropic + vasopressor effects

Therapeutic Uses (2-20 µg/kg/min):

- Clinical use in cardiogenic shock with low cardiac out & decrease renal function
- Increase Glomerular Filtration Rate (GFR)
- $T_{1/2}$: 2 min, onset: 5 min, Duration < 10 min

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2.7 Sympathomimetics

2.7.1 Inotropic sympathomimetics

1. Dopamine hydrochloride	sterile sol	ก
2. Isoprenaline hydrochloride (Isoproterenol hydrochloride)	sterile sol	ก
3. Dobutamine hydrochloride	sterile sol	ข

New drugs acting at dopamine receptors

Mechanism of action:

- D_1 receptors agonist: binds with moderate affinity to α_2 adrenergic receptors

Effects: short-acting intravenous antihypertensive agent with the unique mechanism of action of dopamine1 receptor agonist

Peripheral and renal arteriole vasodilatation

Use: administered using a calibrated infusion pump (0.01 to 1.6 μg/kg/min)

Rapidly acting vasodilator used for not more than 48 h for control of severe hypertension in hospitalized patients

Adverse effects: headache, flushing, dizziness, tachycardia or bradycardia

Fenoldopam

TABLE 3. COMPARISON OF THE RECEPTOR ACTIVITIES OF FENOLDOPAM AND DOPAMINE.

RECEPTOR	LEVEL OF ACTION	
	DOPAMINE*	FENOLDOPAM
Dopaminergic		
DA1	Major	Major
DA2	Major	None
Adrenergic		
α_1	Moderate	None
α_2	Moderate	None
β_1	Major	None

*The receptor-activation profile of dopamine varies according to the dosage. In the low-dose range (2 to 5 μg per kilogram of body weight per minute), only the DA1 and DA2 receptors are activated. At doses of 5 to 10 μg per kilogram per minute, β_1-adrenergic receptors are also activated. In the highest dose range (10 to 50 μg per kilogram per minute) the α-adrenergic receptors are also activated.

N Engl J Med 2001; 345:1548-57

Exercise for reviewing the drugs in non-selective sympathomimetic group

Match the drug and clinical use

α_1 α_2-oxymetazoline
β_1 β_2-isoproterenol
α_1 α_2 β_1 β_2-epinephrine
α_1 α_2 β_1-norepinephrine

A. Topical nasal decongestant

B. **Use for anaphylactic shock**

C. Use for Inotropic agents

D. Vasoconstrictor

มหาวิทยาลัยมหิดล
คณะเภสัชศาสตร์

เกี่ยวกับเรา ▾ หลักสูตร/การเข้าศึกษา ▾ บริการวิชาการ ▾ ผลงานวิจัย ▾ นักศึกษา/บุคลากร ▾

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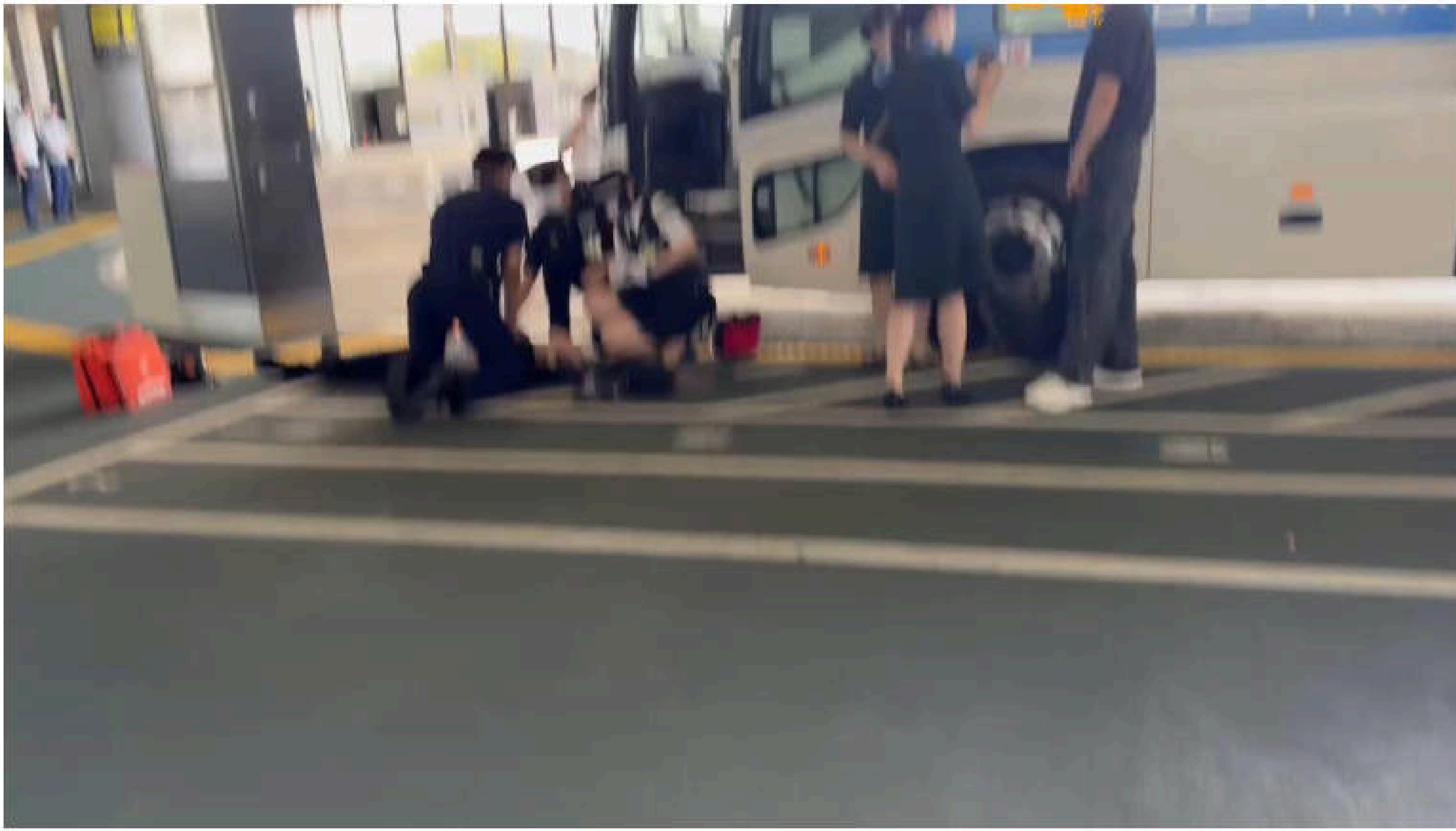
บทความเผยแพร่ความรู้สู่ประชาชน

"อาการตาแดง" (Conjunctivitis)

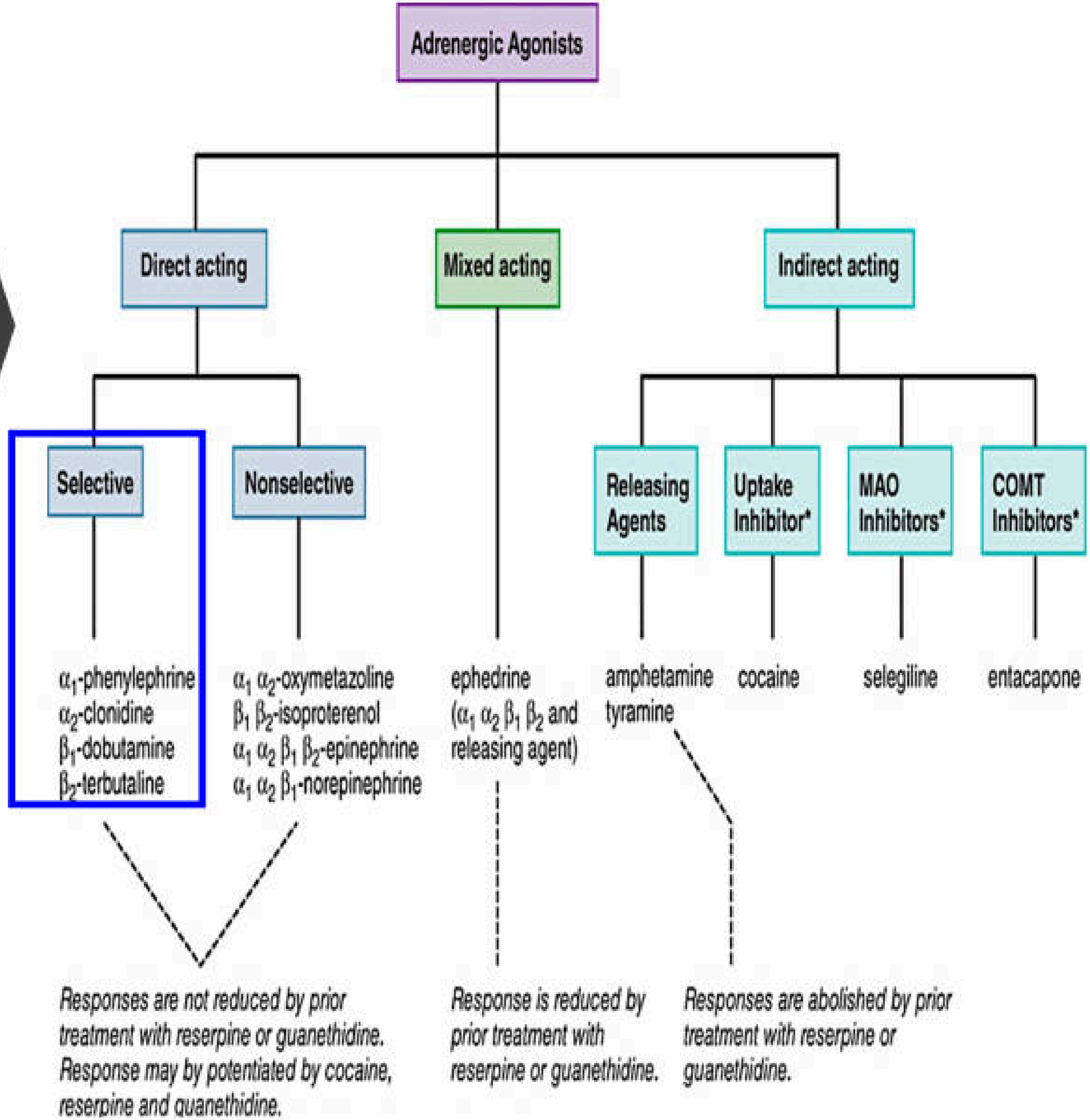
ภาพจาก : http://netdoctor.cdnds.net/15/45/980x490/landscape-1445715811-conjunctivitis-175423329.jpg

1. ยาที่ใช้?
2. Mechanism of action ของยา

Which drug is appropriate for this situation?



Selective
irect acting sympathomim etic drugs



1. α_1-selective adrenergic receptor agonists

Phenylephrine

- **Effects:**
- **1. Smooth muscle:** contraction
- **2. CVS:** vasoconstriction
- 3. **Eyes**: mydriasis
- **Indications:**

1. Nasal decongestant

2. Mydriasis agent (ophthalmic formulation)

3. Vasopressor: dose 0.5-8 µg/kg/min

- **ADRs**: hypertension, Rebound nasal congestion, myocardial infraction (MI)

- **Midodrine**
- orally effective α_1 receptor agonist
- Prodrugs: *desglymidodrine* (active metabolite)
- $T_{1/2}$ of desglymidodrine: 3 h, duration of action: 4–6 h.
- **Uses:** Treatment of orthostatic hypotension
- **ADRs:** Supine hypertension (avoid bedding time)

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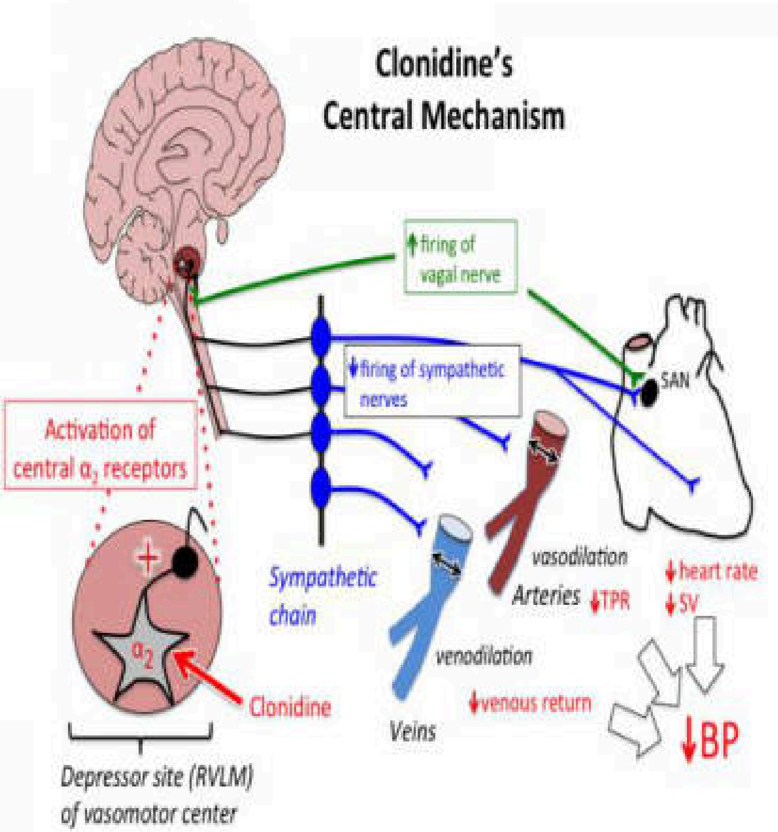
11.3 Mydriatics and cycloplegics

1.	Atropine sulfate	eye drop	ค
2.	Cyclopentolate hydrochloride	eye drop	ค
3.	Phenylephrine hydrochloride	eye drop	ค
4.	Tropicamide	eye drop	ค

2.7.2. Vasoconstrictor sympathomimetics				
1	Norepinephrine	Noradrenaline	sterile sol (as bitartrate or hydrochloride)	ก
2	Ephedrine hydrochloride วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2		sterile sol	ค
3	Midodrine hydrochloride ยากำพร้า		tab	ค
4	Phenylephrine hydrochloride		sterile sol (50 mcg/ml)	ค

2. α_2-selective adrenergic receptor agonists



- **Drugs: Clonidine, methyldopa, guanfacine, guanabenz**
- Decrease blood pressure/ useful in the treatment of hypertension
- **ADR**: CNS side effects: sedation, dizziness, dry mouth, hypotension, syncope, rebound hypertension

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4.10.3 Drugs used in opioid dependence

1.	Clonidine hydrochloride	tab	ข

เงื่อนไข

ใช้สำหรับ heroin withdrawal

2.5.2. Centrally acting antihypertensive drugs			
1	Methyldopa	tab	ก

Suddenly discontinue of Clonidine cause rebound hypertension

New drugs: Brimodinine?

Quiz for review of drugs acting as selective alpha receptor agonist

1. Which drug used for nasal decongestant?
2. Which drug centrally act at brain used for pregnant woman with hypertension?
3. Which drug used for treatment of orthostatic hypotension ?
4. Which drug used for treatment of opioid dependence?

A. Clonidine

B. Phenylephrine

C. Methyldopa

D. Midodrine

3. β-adrenergic receptor agonists

Non selective β **adrenergic receptor** agonists	• Isoproterenol
3.1 Selective β1-adrenergic receptor agonists	• Dobutamine
3.2 Selective β2-adrenergic receptor agonists	• Salbutamol, terbutaline
3.3 β3-adrenergic receptor agonist	• Mirabegron

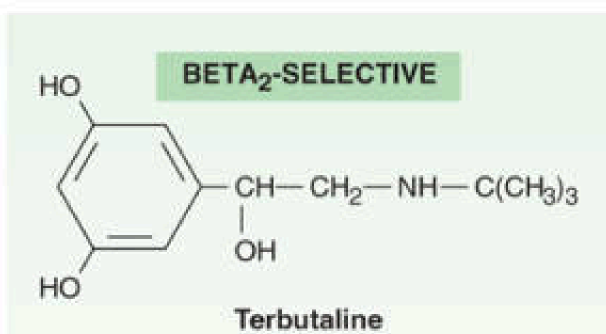
3.1 Dobutamine

- **Selective β1-adrenergic receptor agonists**
- **Effects:** positive inotropic
- **Pharmacokinetics**: not orally absorbed, short $T_{1/2}$ (2 min), rapid onset (1-2 min), duration 10 min
- Initial dose: 0.5-1.0 μg/kg/min therapeutic dose: 2-20 μg/kg/min
- ADR: hypertension, myocardial infarction, hyperglycemia
- **Therapeutic uses: Inotropic sympathomimetic agent**
  - increase cardiac output in post-cardiac surgery, acute congestive heart failure (CHF), acute myocardial infarction
  - Long-term use may cause tachyphylaxis

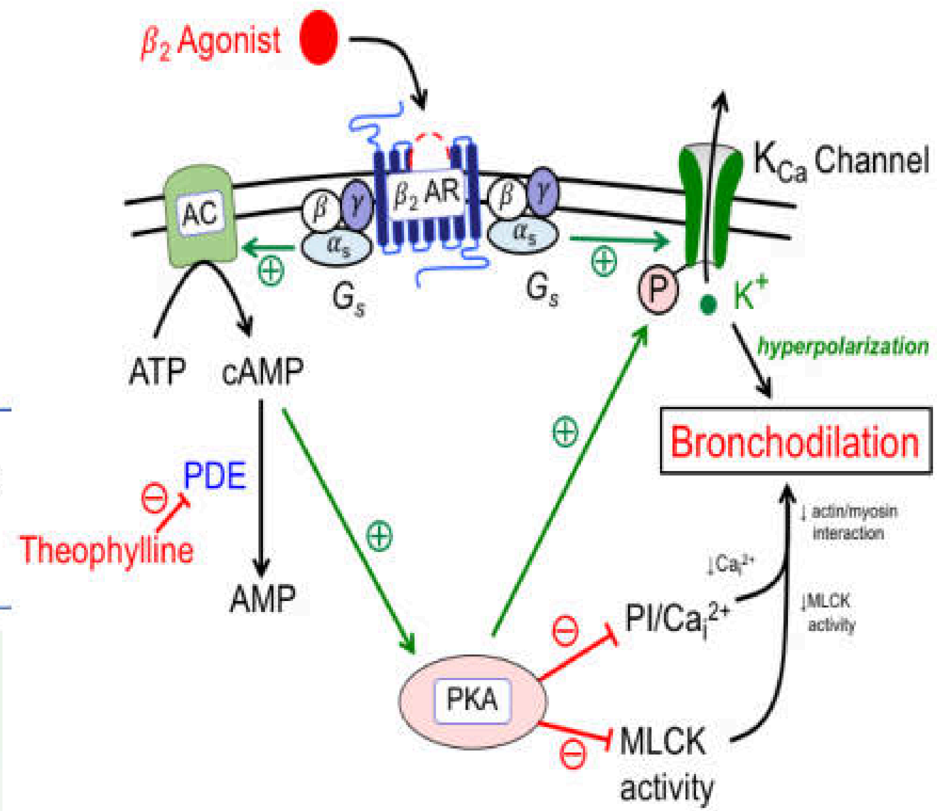
3.2 Selective β_2-adrenergic receptor agonists

Selective Beta$_2$- agonists

- Short-Acting β2 Adrenergic Agonists (**SABA**)
- **Long-Acting β_2 Adrenergic Agonists** (**LABAs**)
- Very Long-Acting β_2 Adrenergic Agonists (**VLABAs**)



Bronchodilator mechanisms for beta-2 agonists



β_2-adrenergic receptor agonists

Pharmacological effects

Smooth muscle:

- smooth muscle relaxation
- Bronchodilatation
- Uterine relaxation

CVS: High dose can stimulate beta1 adrenergic receptor

Decrease mediators from mast cells: leukotriene, histamine

Indications: Treatment of asthma, COPD

Common adverse Effects

- Tremor
- Tachycardia
- Feelings of restlessness, apprehension and anxiety

3.1 Bronchodilators

3.1.1 Adrenoceptor agonists

1.	Procaterol hydrochloride	syr	ก
2.	Salbutamol sulfate	tab, aqueous sol, DPI, MDI, sol for nebulizer	ก
3.	Terbutaline sulfate	tab, syr, sterile sol	ก
4.	Terbutaline sulfate	sol for nebulizer	ก
5.	Procaterol hydrochloride	tab	ง

Selective β_2-adrenergic receptors agonists

Types	Drugs	Formulations	Properties	Indications
Short-Acting β_2 Adrenergic Agonists (SABA)	**Salbutamol or Albuterol (Ventolin®)**	tab, aqueous sol, DPI, MDI, sol for nebulizer	prompt onset of action	treatment of obstructive airway diseases and asthma
	Terbutaline	tab, syr, sterile sol, sol for nebulizer	-orally or subcutaneously or by inhalation -rapid onset and persist for 3-6 h	-long-term treatment of COPD and acute bronchospasm -treatment of status asthmaticus
	Procaterol	Syr, tab		treatment of asthma
	Fenoterol	Inhaler (with ipratropium)	prompt onset of action, and its effect for 4–6 h.	
Long-Acting β2 Adrenergic Agonists (LABAs)	**Salmeterol (not use alone)**	Inhaler (with fluticasone)	Slow onset (not suitable monotherapy) for acute attacks of bronchospasm	agents of choice for nocturnal asthma
	Formoterol (not use alone)	Inhaler (with Budesonide)	Long action (>12 h)	treatment of asthma and bronchospasm, prophylaxis of exercise-induced bronchospasm, and COPD
Very Long-Acting β_2 Adrenergic Agonists (VLABAs)	**Indacaterol**	oral inhalation	once-daily, fast onset of action, combination with tiotropium bromide	COPD
	Vilanterol	oral inhalation	combination with fluticasone	COPD

Other β_2-Selective Agonists

- **Ritodrine**
- Used for uterine relaxant
- PK: rapidly but incompletely (30%) absorbed following oral administration
- May not have clinically significant benefits on perinatal mortality and may actually increase maternal morbidity
- In practical use ***terbutaline*** for prevent premature labor

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7.1.2 Myometrial relaxants

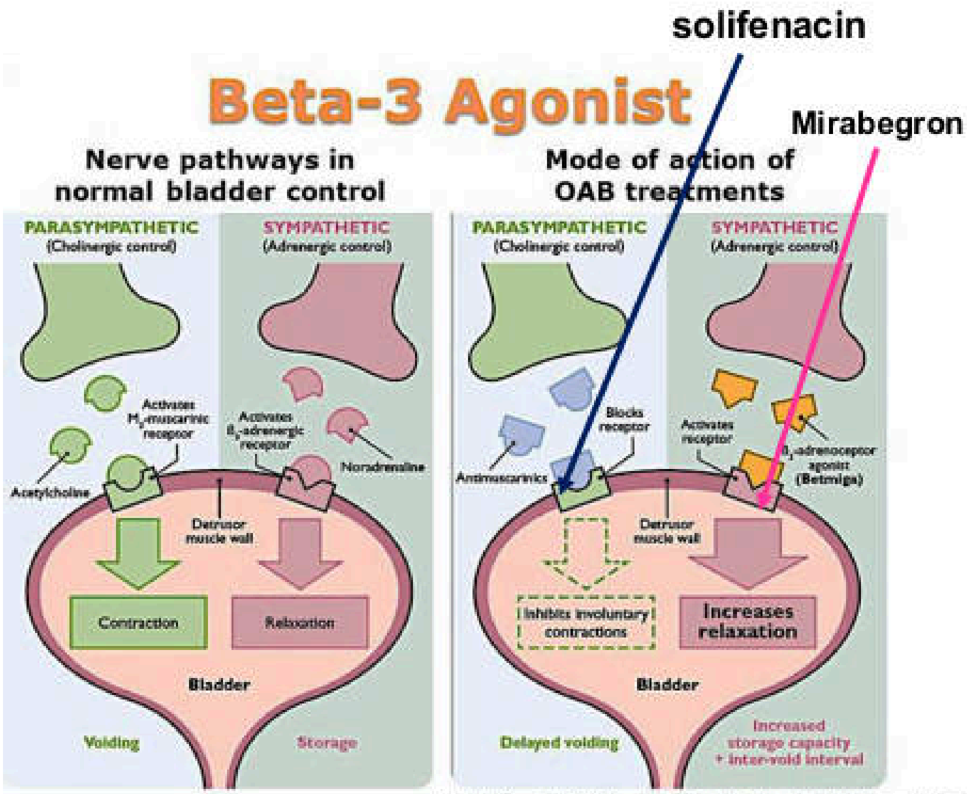
1.	Terbutaline sulfate	tab, sterile sol	ก

สารเร่งเนื้อแดง?

- beta-agonist: Salbutamol, Clenbuteral
- ผลของสารเร่งเนื้อแดงต่อหมู?
  - improved lean **meat**
- อันตรายจากการบริโภคเนื้อหมูที่มีสารเร่งเนื้อแดงตกค้าง?
  - Consuming pork adulterated with salbutamol over a long period has resulted in many symptoms including muscle cramps or tremors, excitement, headache, nausea
- Contraindication:
  - CVS-related diseases, thyroids
- Food hazard: ประเทศไทยและต่างประเทศได้ห้ามใช้สารกลุ่มนี้ในการผลิตอาหารสัตว์โดยเด็ดขาด

3.3 Selective β_3-adrenergic receptor agonists

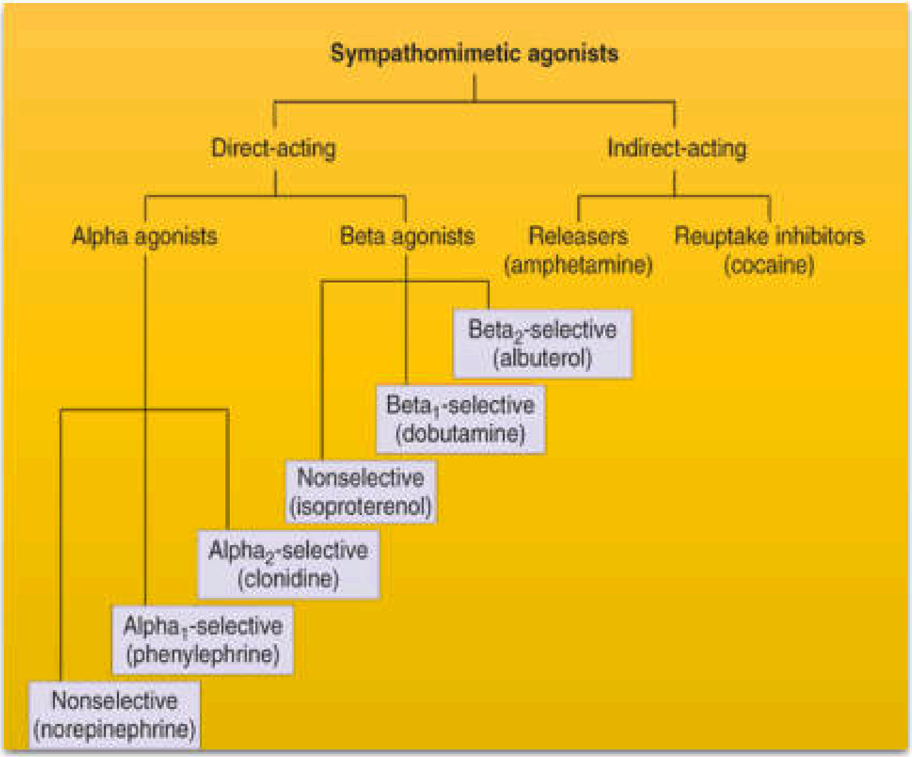
- US-FDA (2018): approved combination of **mirabegron and solifenacin for treatment of** overactive bladder (**OAB)**
- **Expression of β_3 receptor:** brown adipose tissue, gallbladder, and ileum, white adipose tissue and the **detrusor muscle of the bladder**
- **Mirabegron:** β_3 adrenergic receptor agonist approved by US-FDA (2012) for treatment (OAB) and urinary incontinence
- Moderate CYP2D6 inhibitor
- **Side effects**: hypertension, increased incidence of urinary tract infection, headache



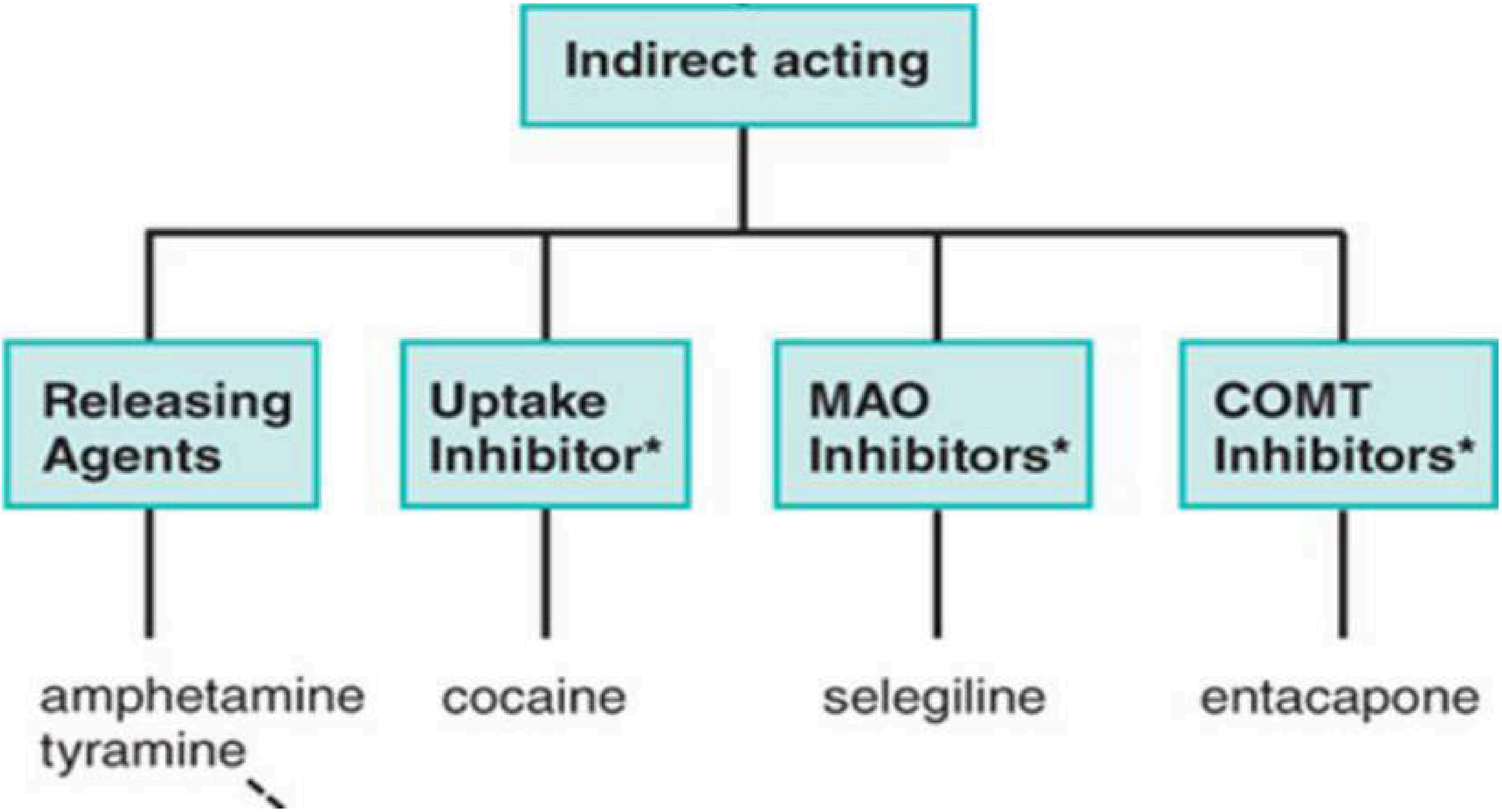
Adapted from Chu FM, Dmochowski R. *Am J Med* 2006;119(3 Suppl 1):3-8.

Summary of sympathomimetic agonists with relative receptor affinities

Drugs	Relative Receptor Affinities
Alpha agonists	
Phenylephrine, methoxamine	$\alpha_1 > \alpha_2 >>>>> \beta$
Clonidine, methyldopa	$\alpha_2 > \alpha_1 >>>>> \beta$
Mixed alpha and beta agonists	
Norepinephrine	$\alpha_1 = \alpha_2; \beta_1 >> \beta_2$
Epinephrine	$\alpha_1 = \alpha_2; \beta_1 = \beta_2$
Beta agonists	
Dobutamine	$\beta_1 > \beta_2 >>>> \alpha$
Isoproterenol	$\beta_1 = \beta_2 >>>> \alpha$
Albuterol, terbutaline, metaproterenol, ritodrine	$\beta_2 >> \beta_1 >>>> \alpha$
Dopamine agonists	
Dopamine	$D_1 = D_2 >> \beta >> \alpha$
Fenoldopam	$D_1 >> D_2$

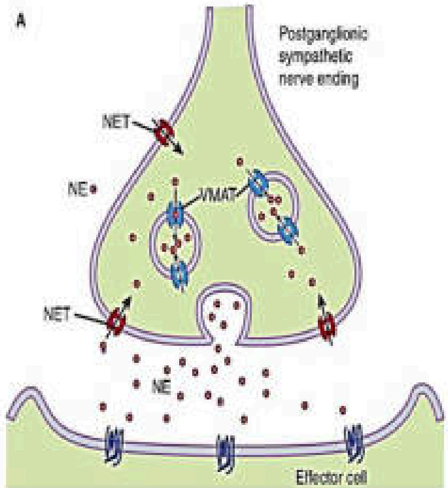


INDIRECT-ACTING SYMPATHOMIMETICS

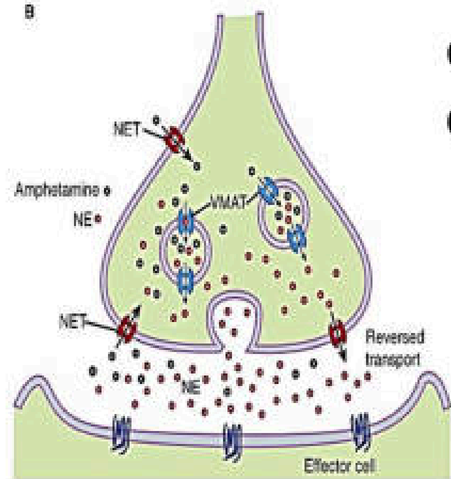


INDIRECT-ACTING SYMPATHOMIMETICS

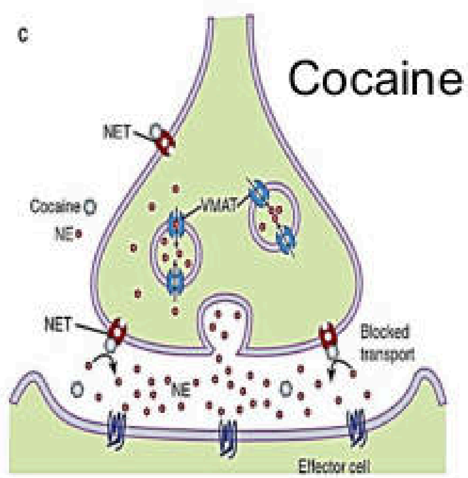
monoamine transporters



amphetamine



Cocaine



- 1. Enter the sympathetic nerve ending and <u>displace</u> stored catecholamine transmitter
  - Amphetamine-like or "displacers"
  - CNS stimulants: <u>amphetamine, Methylphenidate</u>
- 2. <u>Inhibit the reuptake</u> of released transmitter by interfering with the action of the NET
  - Catecholamine Reuptake Inhibitors: <u>Cocaine, Atomoxetine</u>

Mixed-acting sympathomimetic agents

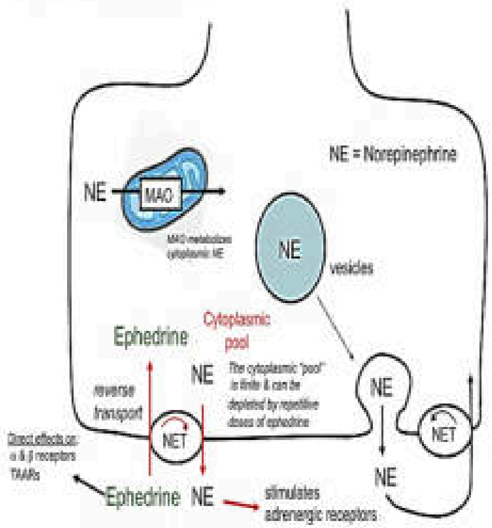
• Ephedrine

- high bioavailability and a relatively long duration
- access to the CNS: mild stimulant
- US-FDA: **banned** the sale of ephedra-containing dietary supplements because of safety concerns

Pseudoephedrine:

- Use as OTC for decongestant mixtures
- Some people used as a precursor in illicit manufacture of methamphetamine

Ephedrine Mechanism



3.7. Systemic nasal decongestants			
1	Pseudoephedrine hydrochloride วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2	tab, syr	ก

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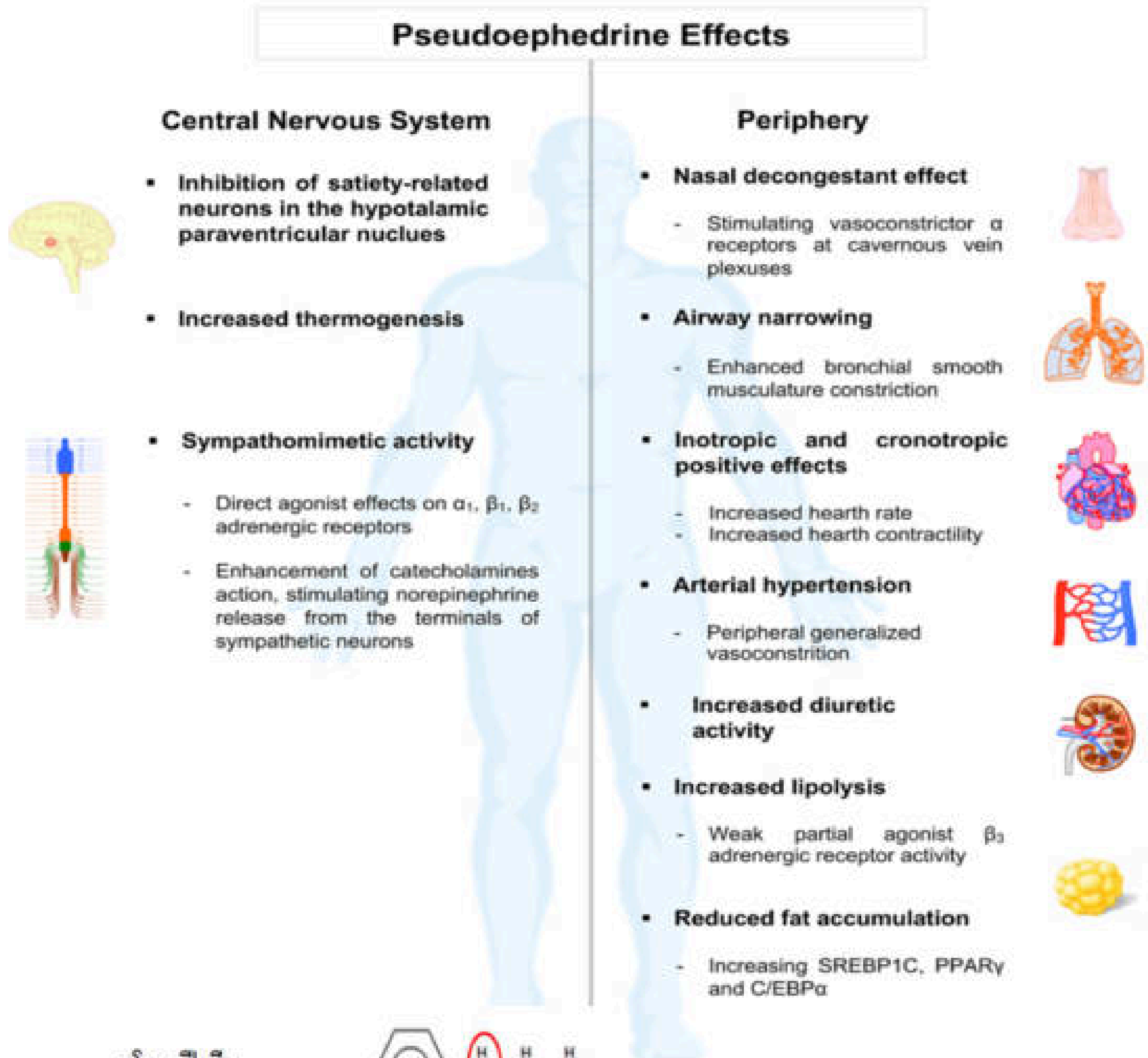
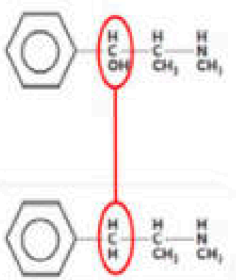
2.7.2. Vasoconstrictor sympathomimetics				
1	Norepinephrine	Noradrenaline	sterile sol (as bitartrate or hydrochloride)	ก
2	Ephedrine hydrochloride วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2		sterile sol	ค
3	Midodrine hydrochloride ยากำพร้า		tab	ค
4	Phenylephrine hydrochloride		sterile sol (50 mcg/ml)	ค

12.2.2 Topical nasal decongestants

1.	Ephedrine hydrochloride *วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2*	nasal drop (hosp) (เฉพาะ 0.5-3%)	ก
2.	Sodium chloride	sterile sol (for irrigation) (เฉพาะ 0.9%)	ก
3.	Oxymetazoline hydrochloride	nasal drop, nasal spray	ข



ซูโดเอฟีดรีน
(**pseudoephedrine**)



เมธแอมเฟตามีน
(**methamphetamine**)

It is possible to convert pseudoephedrine into methamphetamine, so people can potentially use this medication for illicit use

Contraindications of pseudoephedrine?

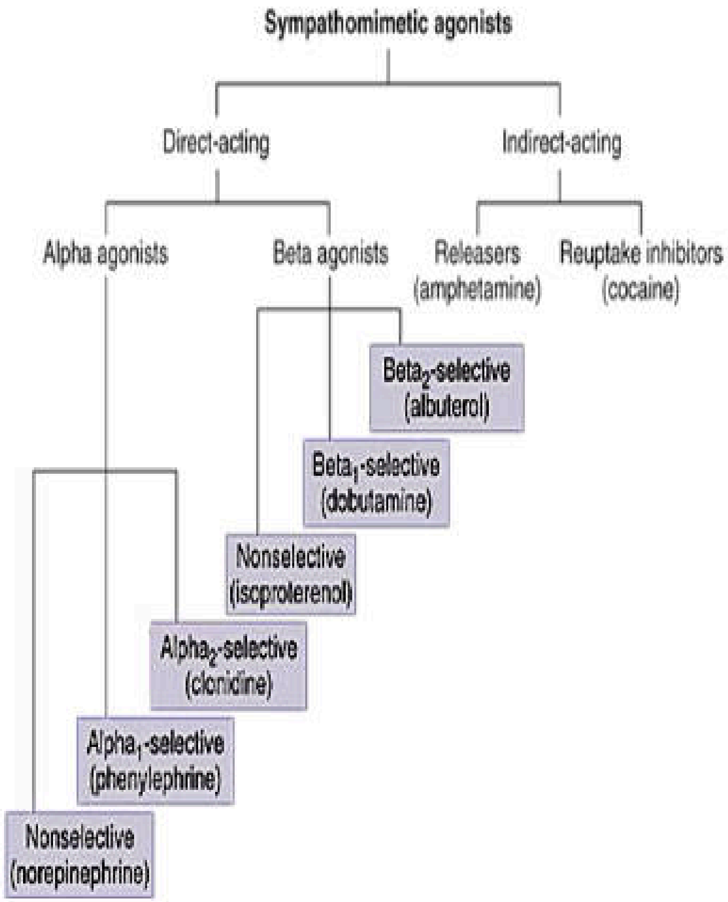
- hypersensitivity to the drug
- cardiovascular diseases (hypertension and coronary artery disease)
- Impaired function of organs responsible for the elimination of the drug
- Hyperthyroidism
- narrow-angle glaucoma
- benign prostatic hyperplasia
- Treatment with monoamine oxidase inhibitors (MAO inhibitors) currently or in the last two weeks
- pregnancy and lactation, and age under 2 years.

Table 3: Over-the-Counter Cough and Cold Medications*

Medication	Side Effects	Interactions	Cautions
Anticholinergic antihistamines	Cognitive dysfunction Hallucinations Sleep disruptions Dry mouth Constipation Impaired diaphoresis Increased thirst Papillary dilation Urinary retention Seizures Arrhythmia Heat stroke	Antiparkinsonian drugs Tricyclic antidepressant agents Phenothiazine medications Oxybutynin	Closed-angle glaucoma Dementia
Pseudoephedrine	Hypertension Vasospasm Arrhythmia Stroke Seizures Hallucinations Chronic headaches Insomnia Tremor	Beta-blockers Digoxin Monoamine oxidase inhibitors	Heart disease Hypertension Thyroid disease Diabetes mellitus Prostatic hypertrophy Urinary retention Closed-angle glaucoma
Topical nasal decongestants	Rhinitis medicamentosa Hypertension	Same as for pseudoephedrine	Same as for pseudoephedrine

*Only drugs with significant risk are listed.
Source: Rolita L and Freedman M, 2008.[1]

Summary of the therapeutic uses of SYMPATHOMIMETIC DRUGS



1. **Cardiovascular Applications**
   - Treatment of acute hypotension, shock, chronic orthostatic hypotension, cardiac arrest: **epinephrine, NE, dopamine, dobutamine, Phenylephrine**
   - Inducing local vasoconstriction: **epinephrine (1:200,000**)

2. Mucous membrane decongestants: phenylephrine, oxymetazoline

3. **Pulmonary Applications:** salbutamol, procaterol, terbutaline

4. Anaphylactic shock: epinephrine (0.3–0.5 mL of a 1:1000 epinephrine solution)

5. **Ophthalmic Applications (glaucoma and mydriatic agents):** : Phenylephrine, α_2-selective agonists (**brimonidine)**

6. Genitourinary applications: β_2-selective agents (eg, **terbutaline**) relax the pregnant uterus

7. Others:

- Hypertension (clonidine, methyldopa)
- Diminishing craving for narcotics and alcohol during withdrawal: **clonidine**
- Overactive Bladder (OAB): **Mirabegron (+ solifenacin)**

Quiz

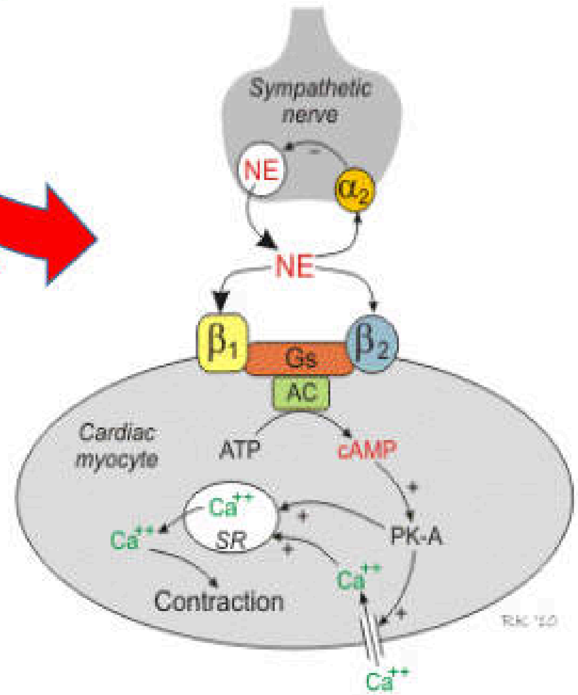
- 1. ยาชนิดใดใช้สำหรับการรักษาภาวะ emergency shock
  - **(A)** Ephedrine
  - **(B)** Epinephrine
  - **(C)** Isoproterenol
  - (D) Norepinephrine

- 2. ยาชนิดใดมีกลไกการออกฤทธิ์ selective β_2 agonists ใช้รักษาโรคหืด
  - **(A)** Terbutaline
  - **(B)** Phenylephrine
  - **(C)** Epinephrine
  - **(D)** Dobutamine

- 3. ยาชนิดใดใช้รักษาภาวะ nasal congestion
  - **(A)** Clonidine
  - **(B)** Phenylephrine
  - **(C)** Dopamine
  - **(D)** Dobutamine

- 4. ยาชนิดใดใช้รักษา hypertension by decrease sympathetic outflow ในสมอง
  - **(A)** Clonidine
  - **(B)** Fenoldopam
  - **(C)** Dopamine
  - **(D)** Salbutamol

ADRENERGIC RECEPTOR ANTAGONISTS

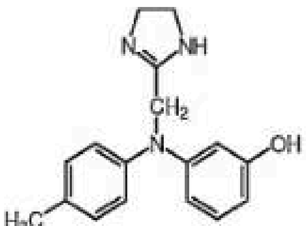


Abbreviations: NE, norepinephrine; Gs, G-stimulatory protein; AC, adenylyl cyclase; PK-A, cAMP-dependent protein kinase; SR, sarcoplasmic reticulum

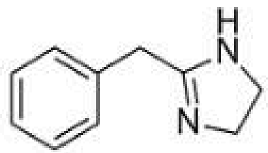
ADRENERGIC RECEPTOR ANTAGONISTS

- Adrenergic Receptor Antagonists
  - α Receptor Antagonists
    - **Nonselective**
      - phenoxybenzamine
      - phentolamine
    - **α_1-selective**
      - prazosin
      - terazosin
      - doxazosin
      - alfuzosin
      - tamsulosin
      - indoramin
      - urapidil
      - bunazosin
    - **α_2-selective**
      - yohimbine
  - β Receptor Antagonists
    - **Nonselective** *(First Generation)*
      - nadolol
      - penbutolol
      - pindolol
      - propranolol
      - timolol
      - sotalol
      - levobunolol
      - metipranolol
    - **β_1-selective** *(Second Generation)*
      - acebutolol
      - atenolol
      - bisoprolol
      - esmolol
      - metoprolol
    - **Nonselective** *(Third Generation)*
      - carteolol
      - carvedilol*
      - bucindolol
      - labetalol*
    - **β_1-selective** *(Third Generation)*
      - betaxolol
      - celiprolol
      - nebivolol

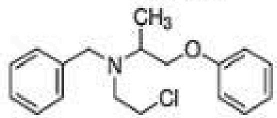
Alpha-receptor antagonists



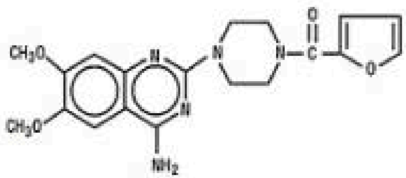
Phentolamine



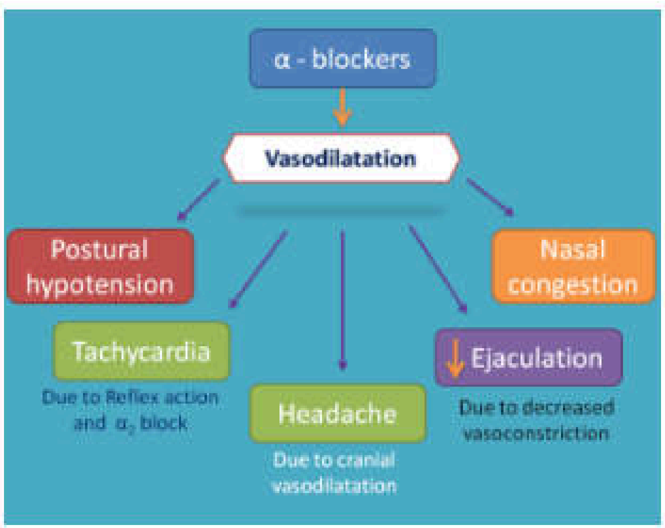
Tolazoline



Phenoxybenzamine



Prazosin



Selective alpha1 blockers: Benign prostatic hypertension (BPH), hypertension	
Alfuzosin, doxazosin, prazosin , tamsulosin, terazosin, silodosin	$\alpha_1 >>>> \alpha_2$
Non-selective alpha blockers: Only limited for hypertensive emergency due to pheochromocytoma	
Phentolamine (reversible, rapid and transient effect)	$\alpha_1 = \alpha_2$
Phenoxybenzamine (irreversible, slower onset and a longer-lasting effect)	$\alpha_1 > \alpha_2$
Selective alpha2 blockers	
Yohimbine, Tolazoline	$\alpha_2 >> \alpha_1$

Alpha-receptor antagonists

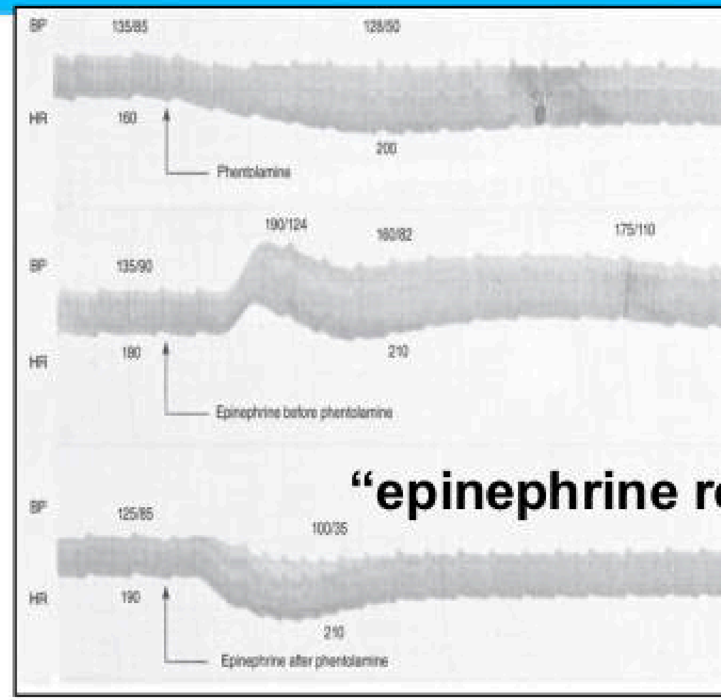
Pharmacologic Effects

1. Cardiovascular Effects

- lowering of peripheral vascular resistance and blood pressure
- Cause: orthostatic hypotension and reflex tachycardia

2. Smooth muscle

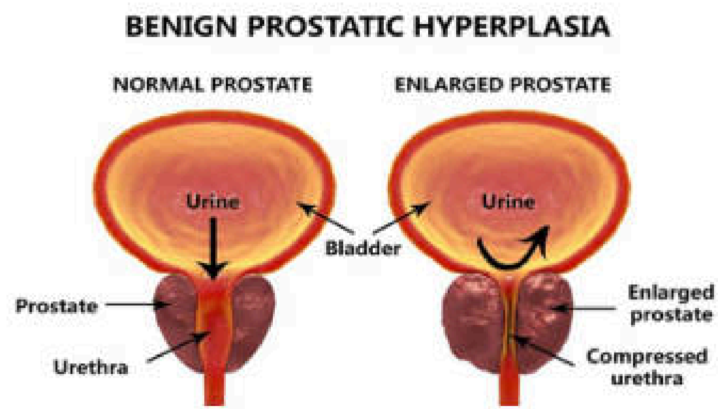
- Relax smooth muscle of bladder and the prostate
- Use for urinary retention due to prostatic hyperplasia



"epinephrine reversal"

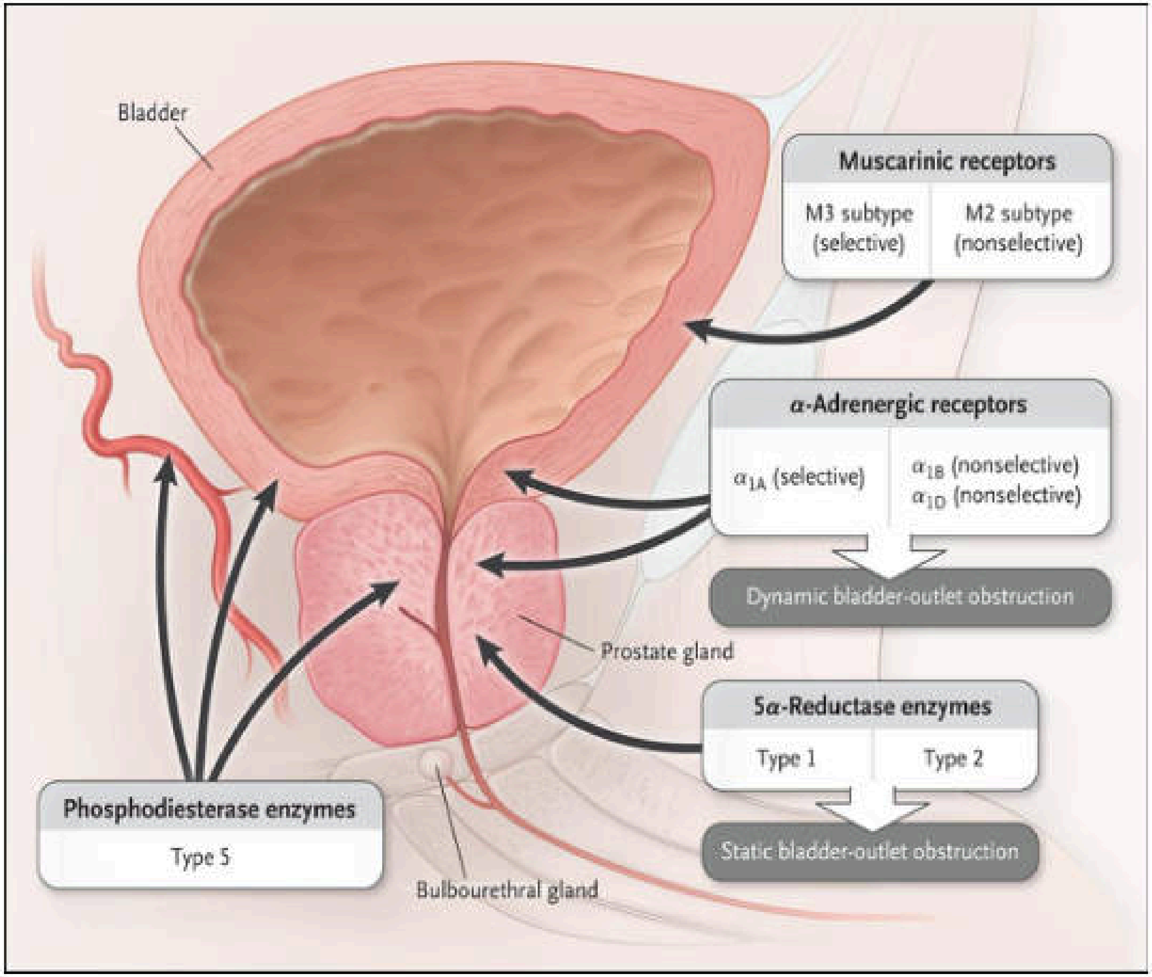
THERAPEUTIC USEs

- **Hypertension**
- **Benign prostatic hyperplasia** (**TAMSULOSIN** in association with FINASTERIDE - 5α-reductase inhibitor, to reduce prostatic volume)
- **Vasospasm** in patients with Raynaud's disease



Adverse effects:
Dizziness, headache, edema, flushing, orthostatic hypotension, first-dose phenomenon

Drugs treatment of Benign prostatic hypertrophy (BPH)



Drugs group	Drugs
Alpha blockers	**Selective**: (α1A-receptor): TamsolusinSildosin
	Non-selective: Prazosin, Doxazosin, terazosin, alfuzosin
5-alpha reductase inhibitors	Finasteride, Dusasteride
Antimuscarinic agents	Selective: solifenacin, Darifenacin
	Nonselective: trospium, oxybutynin, Tolterodine
PDE-5 inhibitors	Tadalafil
Dual-product	Dusasteride-Tamsolusin

N Engl J Med 2012; 367:248-257

Alpha-receptor antagonists (cont.)

2.5.3 Alpha-adrenoceptor blocking drugs

1.	Prazosin hydrochloride	tab	ข
2.	Doxazosin mesilate	immediate release tab	ค

Drugs	Indications	Dosage forms
Alfuzosin (Xatral®)	BPH	ER tab, 10 mg
Doxazosin (Cardura®)	Hypertension , BPH	Tab, 2,4 mg (T1/2: 20 h)
Prazosin (Minipress®)	Hypertension	Tab, 1,2 mg (T1/2: 2-3h)
Tamsolusin (Hernal®)	BPH	Tab, 0.4 mg
Terazosin (Hytrin®)	Hypertension , BPH	Tab, 1,2,5 mg (T1/2: 12h)
Sildosin (Urief®) Selective alpha1A antagonist	BPH	Tab, 4 mg

Yohimbine: α_2-selective antagonist

- Little clinical usefulness
- Treatment of orthostatic hypotension (not common)
- Treat male erectile dysfunction (not common) some country use as supplement

Quiz α- adrenergic receptor antagonist

- 1. Which of the following alpha blocking drugs is non-selective?

A. Tamsolusin

B. Doxazosin

C. Terazosin

D. Phenoxybenzamine

2. Which alpha blocking drug is selective alpha1A-receptor?

A. Prazosin

B. Doxazosin

C. Terazosin

D. Tamulosin

3. What is not the adverse drugs reaction from prazosin?

A. Postural hypotension

B. Hypertension

C. Headache

D. Dizziness

4. Which time should be consider for taking doxazosin?

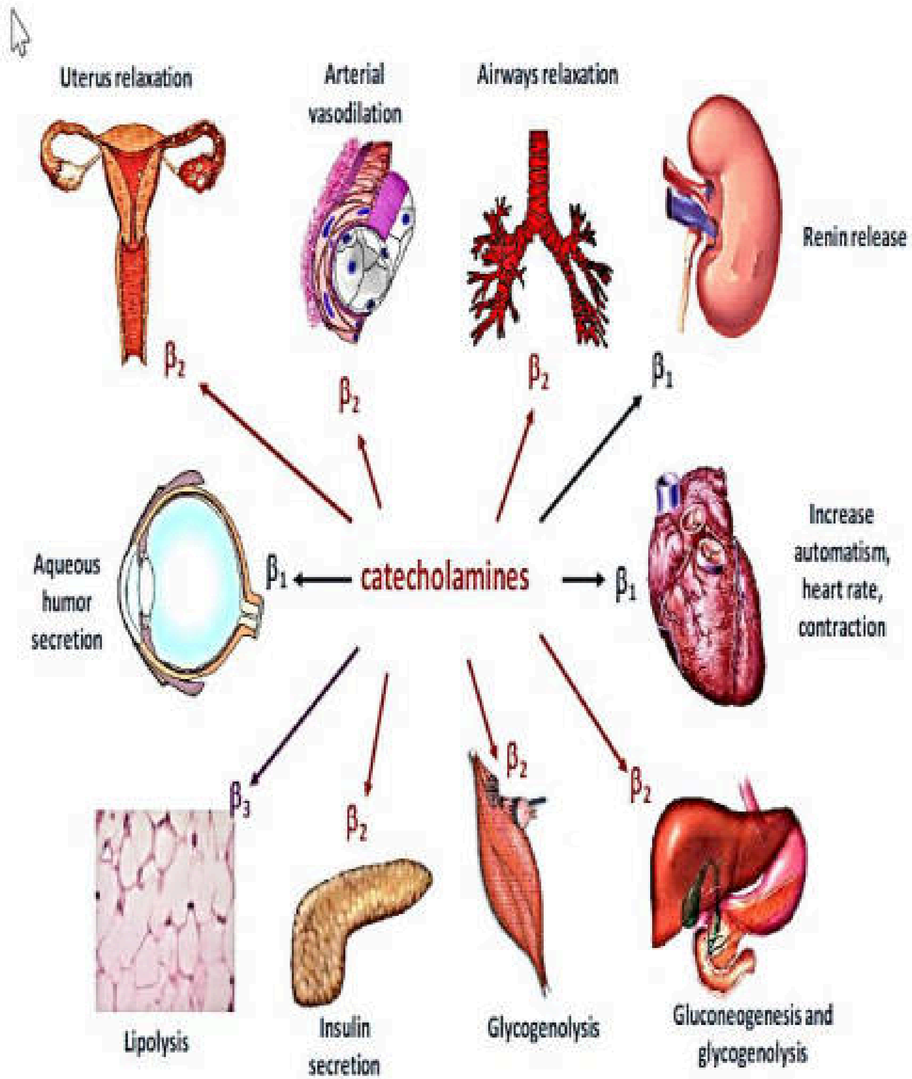
A. Morning

B. Afternoon

C. Evening

D. Before bedding

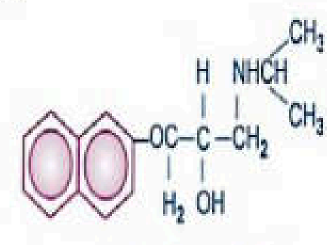
β-ADRENERGIC RECEPTOR LOCALIZATION AND RELATIVE PHYSIOLOGICAL FUNCTION



Beta-blockers

First generation (older, nonselective)	Second generation (β_1 selective)	Third generation (with additional α blocking and/or vasodilator property)
Propranolol Timolol Sotalol Pindolol	Metoprolol Atenolol Acebutolol Bisoprolol Esmolol	Labetalol Carvedilol Celiprolol Nebivolol

The pharmacology of propranolol is described as prototype.



PROPRANOLOL

Beta Adrenergic Antagonists

β Receptor Locations
β 1 = Heart (1 heart)
β 2 = Lungs (2 lungs)

Monitoring
BP & HR

Mortality Benefit for HFrEF
Metoprolol succinate
Carvedilol
Bisoprolol

Subtypes based on Receptor Specificity

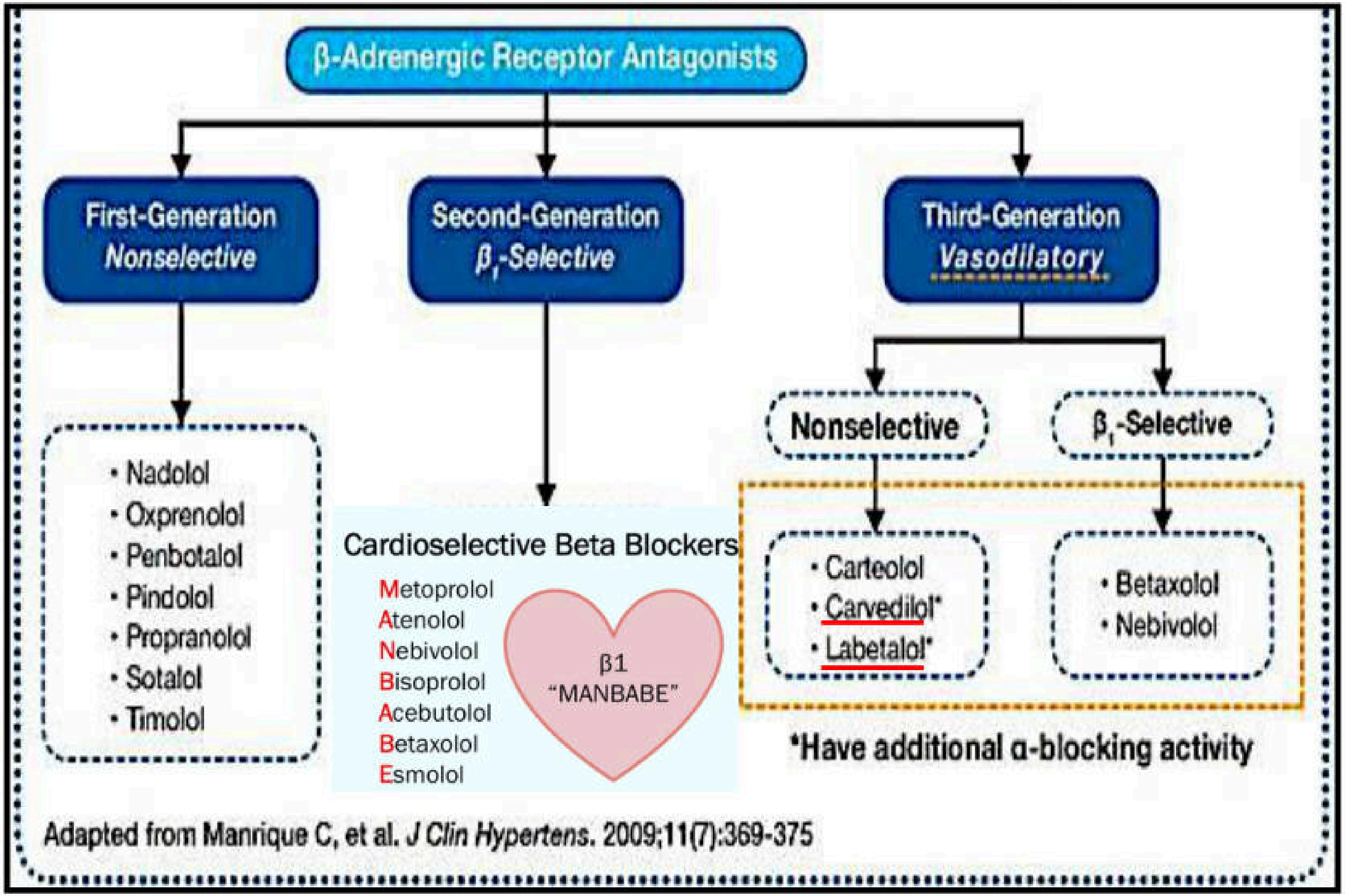
Cardioselective
(B1 selective) = MANBABE
- Metoprolol
- Atenolol
- Nebivolol
- Betaxolol
- Acebutolol
- Bisoprolol
- Esmolol

β1
"MANBABE"

Mixed α/β
Doesn't end in olol
- Carvedilol
- Labetalol

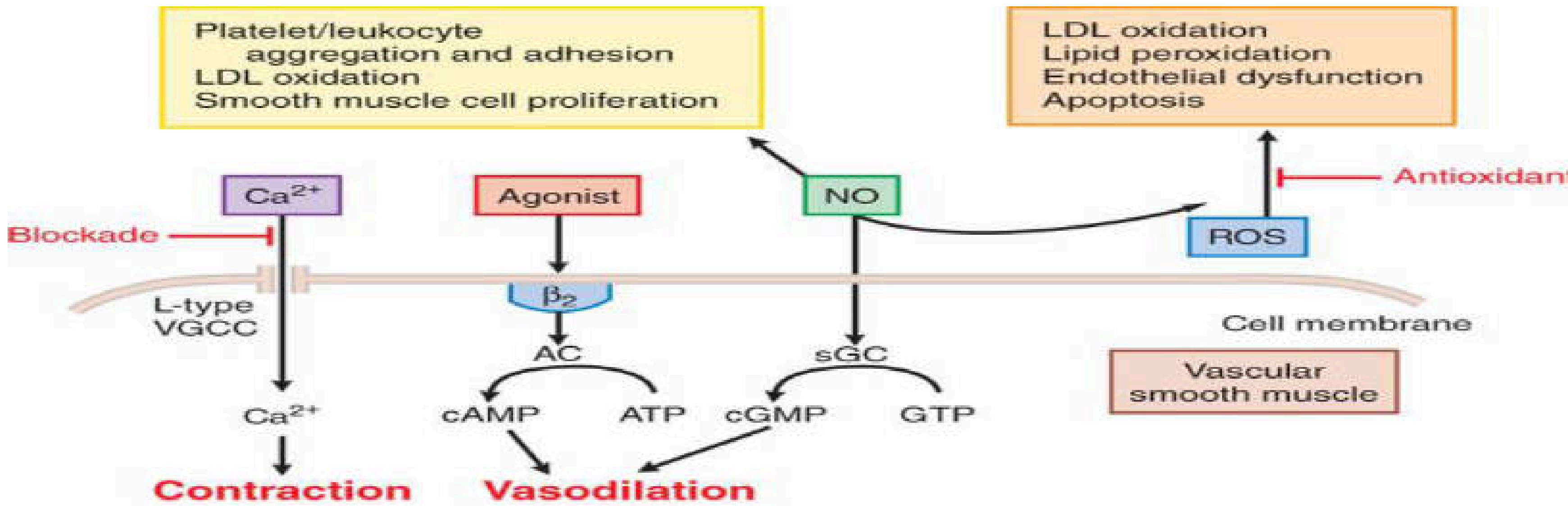
Nonselective
- Propranolol
- Nadolol

Classification of **beta-blockers**



https://specialty.mims.com

Mechanisms underlying actions of vasodilating ß-blockers in blood vessels

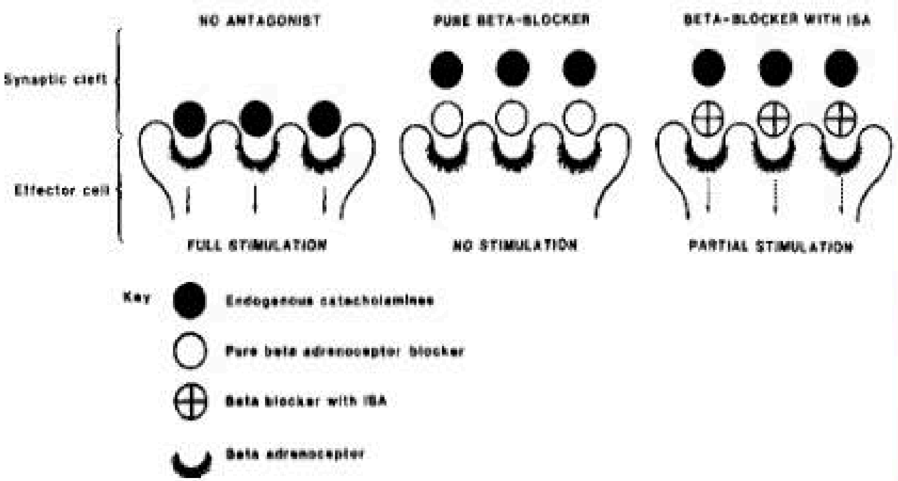


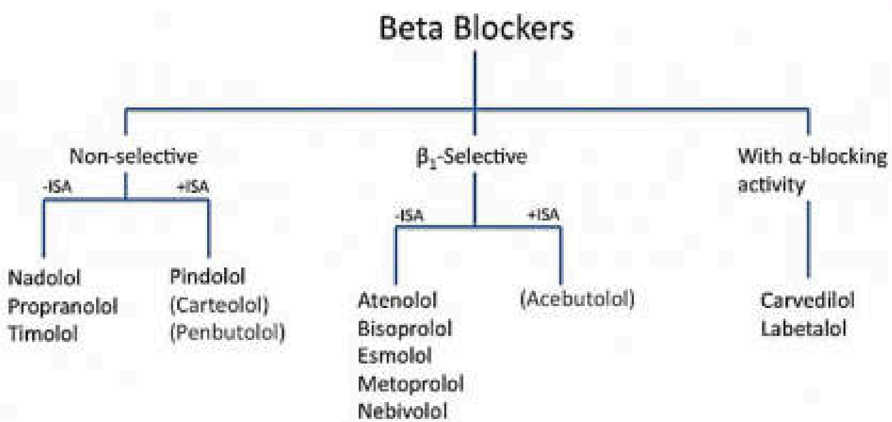
NITRIC OXIDE PRODUCTION	β_2 RECEPTOR AGONISM	α_1 RECEPTOR ANTAGONISM	Ca^{2+} ENTRY BLOCKADE	K^+ CHANNEL OPENING	ANTIOXIDANT ACTIVITY
Celiprolol	Celiprolol	Carvedilol	Carvedilol	Tilisolol	Carvedilol
Nebivolol	Carteolol	Bucindolol	Betaxolol		
Carteolol	Bopindolol	Bevantolol	Bevantolol		
Bopindolol		Nipradilol			
Nipradilol		Labetalol			

AC: adenylyl cyclase; sGC: soluble guanylyl cyclase; NO: nitric oxide; ROS: reactive oxygen species; VGCC: voltage-gated Ca2+ channel

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13ed.

Beta blockers with intrinsic sympathomimetic activity (ISA):
(or with ***partial agonists***)

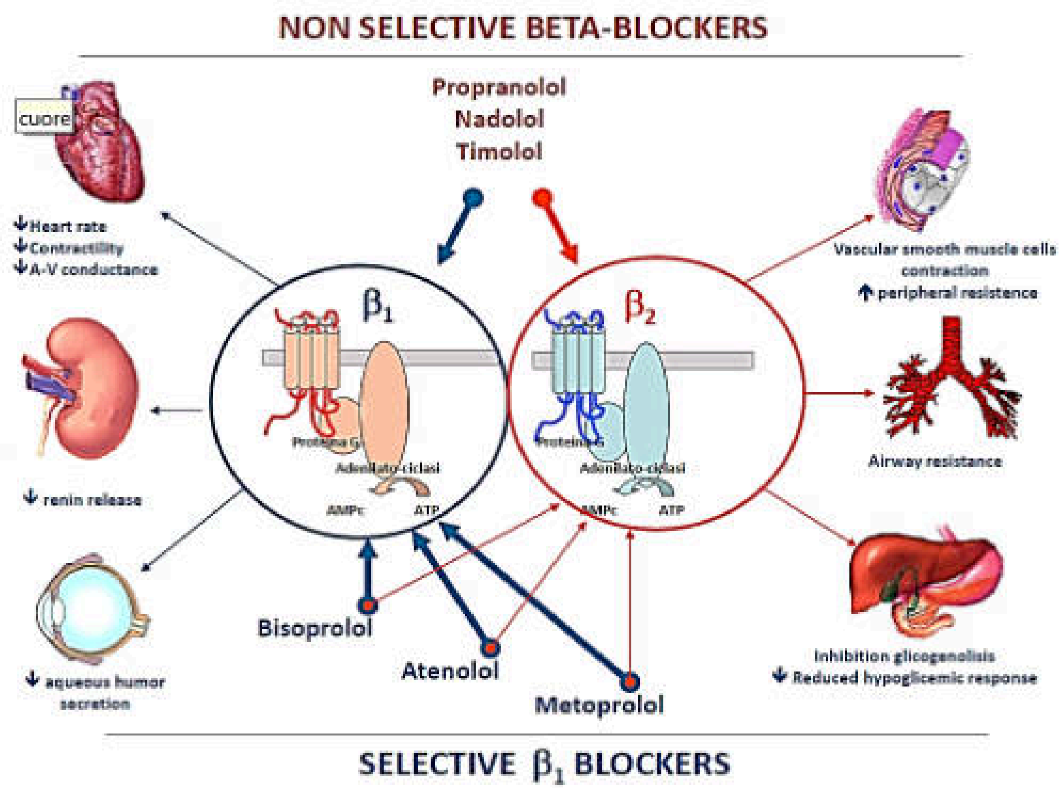




ISA: Intrinsic Sympathomimetic Activity (partial agonists)
(Grey font: Drugs not emphasized at Tulane)

ISA Is Controversially Discussed

Advantage	Disadvantage
1. bradycardia: ISA-blockers could be useful in patients with bradycardia 2. less likely to disturb with — carbohydrate, lipid metabolism — peripheral circulation — lung function	1. Efficacy in CAD — reduction in HR is less compared to β-blockers without ISA→increasing O2 consumption with a deterioration in O2 balance — tendency towards an increase in HR at night (sympathetic tone is low) 2. ISA-blockers seem to be less effective in — secondary prevention — prophylaxis of migraine — hyperthyroidism 3. ISA specific side effects — sleep disturbances — restlessness — nervou — tremor IWAN TANABE



Beta-blocker effects	
β1	**β2**
↓ HR	↑ Airway resistance
↓ AV conduction	Arterial vasoconstriction
↓ Cardiac Output	↓ Gluconeogenesis
↓ O2 comsumption	↓ Glycogenolysis
↓ Blood Pressure	↓ Tremors
↓ Renin	
↓ Aqueous humour	

Effects of beta-blockers on different organs

Organ system	Effect
Heart	Negative inotropic and chronotropic effects
Bronchopulmonary	Increased airway resistance
Kidney	Reduced activity of the renin-angiotensin-aldosterone system
Central and peripheral nervous system	Decreased sympathetic nervous system activity
Metabolic	Inhibition of glycolysis, glucagon secretion and lipolysis
Skeletal muscle	Reduction of exercise capacity
Eye	Increased outflow and reduced secretion of aqueous humor

Pharmacological activities of beta-blockers

System	Effects
1. CVS	Negative inotropic, chronotropic lower blood pressure, treatment of angina and chronic heart failure , myocardial infarction
	Local anesthetic action or "membrane-stabilizing action"
2. Respiratory Tract	increase in airway resistance (beta2 receptor)
3. Eye	• reduce intraocular pressure, especially in glaucoma (reduce aqueous humor production)
4. Endocrine Effects	• inhibit lipolysis, glycolysis, glucagon secretion • inhibit recovery from hypoglycemia • Increase plasma concentrations of VLDL, decrease HDL cholesterol
5. Kidney	• Reduce the activity of renin angiotensin system
6. Other Effects	• Block catecholamine-induced tremor • Block inhibition of mast cell degranulation by catecholamine

Pharmacokinetic Properties of the Beta-Receptor Antagonists

DRUG	MEMBRANE STABILITY	IAA	LIPID SOLUBILITY	% ABSORPTION	% ORAL AVAILABILITY	PLASMA $t_{1/2}$ (h)
			Classical non-selective β blockers: First generation			
Nadolol	0	0	Low	30	30-50	20-24
Penbutolol	0	+	High	~100	~100	~5
Pindolol	+	+++	Low	>95	~100	3-4
Propranolol	++	0	High	<90	30	3-5
Timolol	0	0	Low/moderate	90	75	4
			$β_1$-Selective β blockers: Second generation			
Acebutolol	+	+	Low	90	20-60	3-4
Atenolol	0	0	Low	90	50-60	6-7
Bisoprolol	0	0	Low	≤90	80	9-12
Esmolol	0	0	Low	NA	NA	0.15
Metoprolol	+[a]	0	Moderate	~100	40-50	3-7
			Non-selective β blockers with additional actions: Third generation			
Carteolol	0	++	Low	85	85	6
Carvedilol	++	0	Moderate	>90	~30	7-10
Labetalol	+	+	Low	>90	~33	3-4
			$β_1$-selective β blockers with additional actions: Third generation			
Betaxolol	+	0	Moderate	>90	~80	15
Celiprolol	0	+	Low	~74	30-70	5
Nebivolol	0	0	Low	NA	NA	11-30

IAA, intrinsic agonist activity.

[a]Detectable only at doses much greater than required for β blockade.

Case study

- **Male 25 yrs. old patient** visits his primary care physician for a routine check-up
- **Medical History:** Hypertension (diagnosed 10 years ago)
  - Mild bradycardia (resting heart rate: 55 bpm)
  - Mild hyperlipidemia
  - No history of coronary artery disease
- **Current Medications:**
  - Amlodipine (5 mg daily) for hypertension
  - Simvastatin (20 mg daily) for hyperlipidemia
- **Clinical Presentation:**
  - His blood pressure has been moderately elevated over the past few visits (around 150/90 mmHg).
  - Despite being on amlodipine, his blood pressure is not adequately controlled.

- 1. Which drug is appropriate for the treatment

a. Propranolol

b. Pindolol

c. Acebutolol

2. What is the drug's mechanism of action in answer?

a. Cardioselective beta- blocker

b. Non selective beta-blocker

c. Beta-blocker with ISA

PHARMACOKINETICS

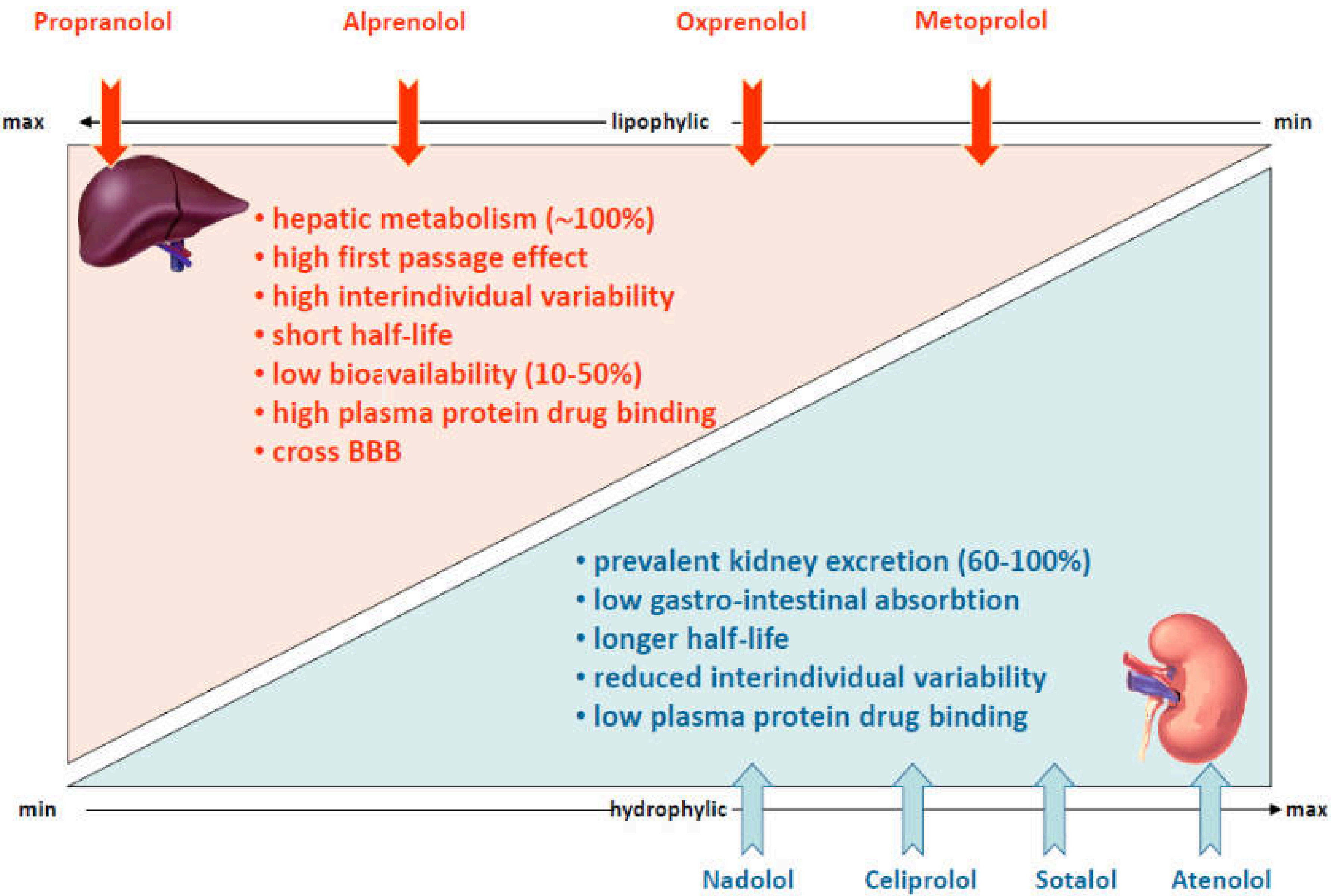


Table 2. Differences in the actions of some beta-blockers.[5]

Drug	Lipid solubility	Cardio-selectivity	Partial agonist activity	Membrane-stabilizing activity	Peripheral vasodilatation
Atenolol	–	+	–	–	–
Bisoprolol	+	+++	–	–	+
Carvedilol	+++	–	–	++	++
Labetalol	+	–	–	±	++
Metoprolol	+	+	–	±	+
Oxprenolol	+	–	+	+	–
Practolol	–	++	+	–	–
Propranolol	++	–	–	++	±
Sotalol	–	–	–	–	–
Timolol	+	–	±	±	+

+ means that the drug has the indicated property, to varying degrees depending on the number of symbols; – means that it does not; ±, it is not clear whether it does.

British Journal of General Practice, *June 2008*

- ยากลุ่ม beta-blocker ชนิดใดควรเลือกใช้ในผู้ป่วยความดันโลหิตสูงที่มีภาวะไขมันในเลือดสูงร่วมด้วย
- ยากลุ่ม beta-blocker ชนิดใดควรเลือกใช้ในผู้ป่วยความดันโลหิตสูงที่มีภาวะ asthma ร่วมด้วย
- ยากลุ่ม beta-blocker ชนิดใดควรเลือกใช้ในผู้ป่วยความดันโลหิตสูงที่มีการทำงานของไตบกพร่อง
- ยากลุ่ม beta-blocker ชนิดใดควรเลือกใช้ในผู้ป่วย cardiac arrhythmia

Adverse Effects

BETA BLOCKER SIDE EFFECTS

"BALD FISH"

B	RONCHOCONSTRICTION RADYCARDIA	bradycardia results in lethargy, fatigue and exercise intolerance
A	RRHYTHMIAS	
L	ETHARGY	
D	ISTURBANCE IN GLUCOSE METABOLISM	causes and masks S/S of hypoglycemia
	Dyslipidemia (↑TG, ↑ LDL, ↓ HDL)	
F	ATIGUE (reduce exercise capacity)	
I	NSOMNIA	
S	EXUAL DYSFUNCTION	
H	YPOTENSION	

- *Others:*
- *Abrupt discontinuation after long-term treatment: exacerbate angina and increase the risk of sudden death*

Over-dosage

- Hypotension
- Bradycardia
- Prolonged AV conduction times
- Widened QRS complexes
- Seizures
- Hypoglycemia
- Bronchospasm

Drug Interactions

Increase efficacy:

- Cimetidine (hepatically metabolized beta-blockers)
- Quinidine (hepatically metabolized beta-blockers)
- Food (hepatically metabolized beta-blockers)

Beta-blockers

Decrease efficacy:

- NSAIDs
- Withdrawal of clonidine
- Agents that induce hepatic enzymes, including rifampin and phenobarbital

Interactions

- Propranolol hydrochloride induces hepatic enzymes to increase clearance of drugs with similar metabolic pathways.
- Beta-blockers may mask and prolong insulin-induced hypoglycemia.
- Heart block may occur with non-DHP- Ca-antagonists.
- Sympathomimetics cause unopposed alpha-adrenoceptor mediated vasoconstriction.
- Beta-blockers increase angina-inducing potential of cocaine.

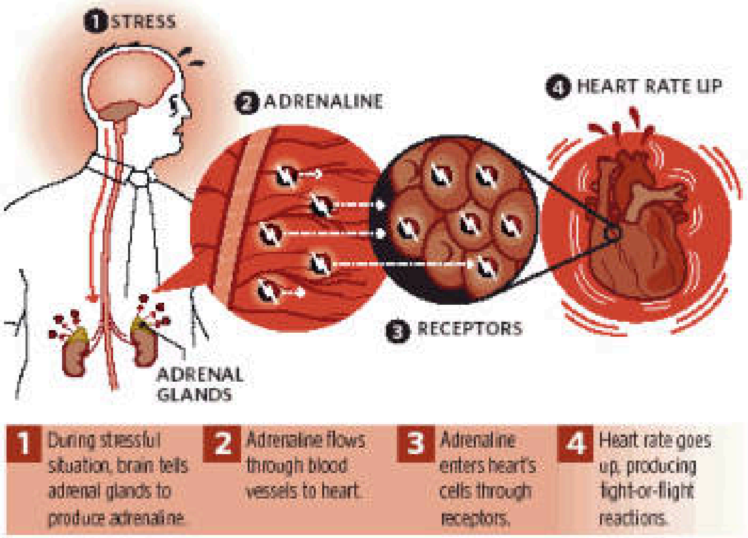
Contraindications

1. Asthma, COPD
2. Prinzmetals angina
3. Bradycardia

   Heart Block

   Acute decompensated heart failure
4. Peripheral Vascular disease

CLINICAL PHARMACOLOGY OF THE BETA-RECEPTOR–BLOCKING DRUGS



- **CVS:**
  - **Hypertension, Ischemic Heart Disease,**
  - **Cardiac Arrhythmias, Heart Failure, pheochromocytoma (↓Epi, NE)**
- **Non- CVS uses:**
  - **Glaucoma (↓ aqueous humor):** Timolol, betaxolol
  - **Hyperthyroidism:** Propranolol
  - **Neurologic Diseases:** propranolol prevent migraine headache, stage fright, nervousness, panic
  - **Miscellaneous:**
    - **Esophageal varices** [หลอดเลือดขอดในหลอดอาหาร] (Nadolol combination with isosorbide mononitrate)
    - **Infantile hemangioma** [โรคเนื้องอกหลอดเลือด](propranolol 2 mg/kg/d)

ยา	รูปแบบ	บัญชียา	เงื่อนไข
Beta-adrenoceptor blocking drugs			
1.Atenolol	tab	ก	
2. Metoprolol tartrate	tab immediate release	ก	
3.Propranolol hydrochloride	tab	ก	
4.Carvedilol	tab	ค	ใช้สำหรับ heart failure with reduced ejection fraction
5. Labetalol hydrochloride ยากำพร้า	sterile sol	ค	ใช้สำหรับ hypertensive emergencies
6. Esmolol hydrochloride	sterile sol	ง	1.ใช้สำหรับรักษาภาวะที่หัวใจมีการเต้นเร็วผิดปกติ (supraventricular tachycardia, non-compensatory sinus tachycardia) ในผู้ป่วยที่มีความเสี่ยงต่อการเกิดภาวะแทรกซ้อนของหัวใจและหลอดเลือดทั้งในระหว่างและหลังการผ่าตัด 2. ใช้สำหรับควบคุมการเต้นของหัวใจให้ช้ากว่าปกติ หรือไม่ให้เต้นเร็ว ทั้งในระหว่างและหลังการผ่าตัด หรือในระหว่างการระงับความรู้สึก หรือ หาหัตถการ เช่น การตรวจวินิจฉัย computed tomography (CT) heart เป็นต้น 3. ใช้โดยวิสัญญีแพทย์ ใช้ยานี้ในระยะสั้นไม่เกิน 24 ชั่วโมง
Drugs for treatment of glaucoma			
Timolol maleate	eye drop	ก	
Drugs used in the prophylaxis of migraine			
Propranolol hydrochloride	tab	ก	
Drugs used in movement disorders			
Propranolol hydrochloride	tab	ก	ใช้สำหรับ essential tremor

Quiz beta-blockers

- 1. Which of the following statements about beta-blockers is false?

A. Beta-blockers increase renin secretion

B. Non-selective beta-blockers are contraindicated in asthma

2. Beta-blockers – aside from their cardiovascular indications – are also used in the treatment of which condition?

A. Glaucoma

B. Reynaud's phenomenon

C. Asthma

3. Which is cardio selective beta blocker?

A. Nadolol

B. Propranolol

C. Atenolol

4. Which of the following beta-blockers also has additional alpha-blocking activity?

A. Nadolol

B. Pindolol

C. Labetalol

5. Which of the following beta-blockers has intrinsic sympathomimetic activity (ISA)?

A. Bisoprolol

B. Pindolol

C. Propranolol

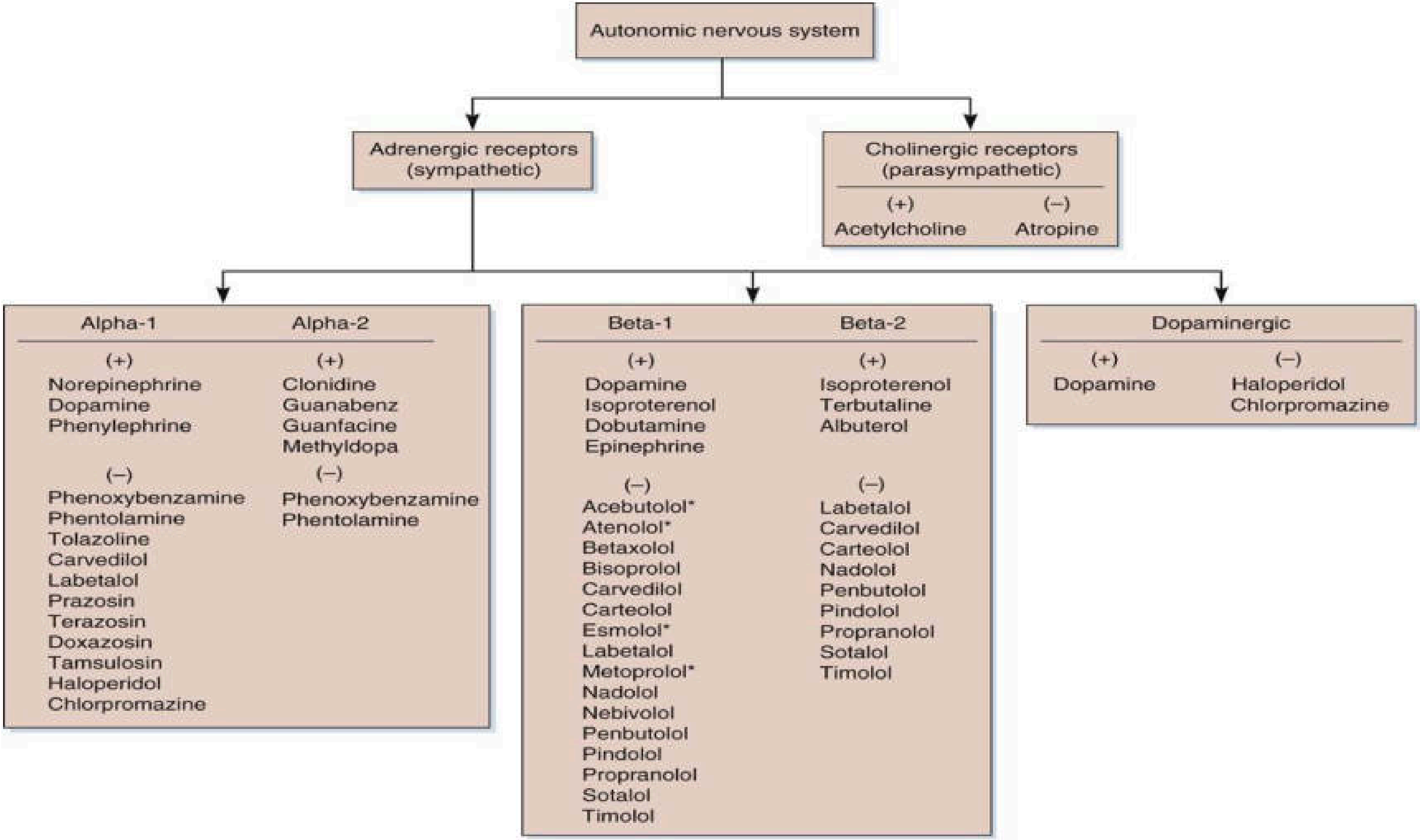
6. Which is not side effect from propranolol?

A. Heart block

B. Increase HDL

C. Fatigue

Summary pharmacology of sympathetic nervous system

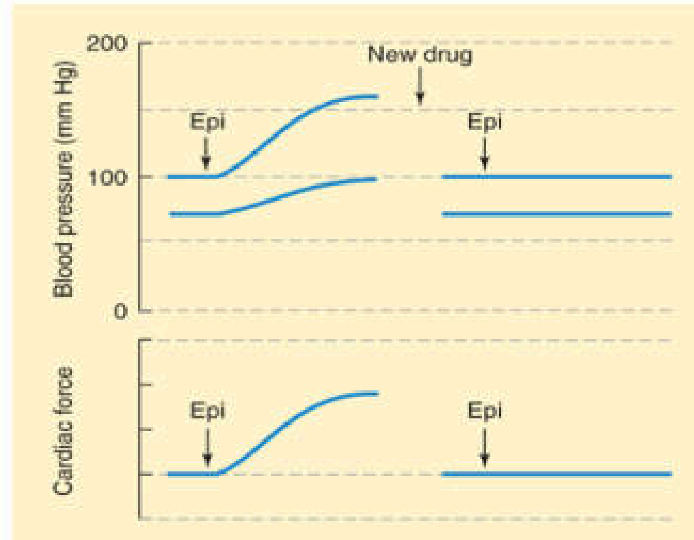


QUIZ

1.A 56-year-old man has hypertension and an enlarged prostate, which biopsy shows to be benign prostatic hyperplasia. He complains of urinary retention. Which of the following drugs would be the most appropriate initial therapy?

- **(A)** Albuterol
- **(B)** Atenolol
- **(C)** Metoprolol
- **(D)** Prazosin
- **(E)** Timolol

2. A new drug was administered to an anesthetized animal with the results shown here. A large dose of epinephrine (Epi) was administered before and after the new agent for comparison.



Which of the following agents does the new drug most closely resemble?

(A) Atenolol
(B) Atropine
(C) Labetalol
(D) Phenoxybenzamine
(E) Propranolol

Thank you

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